Research Proposal

Co-Design of a Fault-Tolerant RISC-V System-on-Chip for Autonomous Closed-Loop Additive Manufacturing in Space Environments

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Proposed duration	12 months
Estimated resources	≈ Rs. 67,200 (US$790), released in four tranches against the review gates
Target technology readiness	TRL 3–4 at completion
Intended outputs	2 peer-reviewed papers; open-source RTL, fault-injection framework and defect dataset

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1 Abstract

3D printers have already worked in space, but they are not autonomous. The polymer printers on the International Space Station (ISS) are run by the crew and commanded from the ground, so a person fixes every problem that comes up [1, 2]. That approach will not work further out. A printer on the Moon, on a free-flying satellite, or on a deep-space mission has no crew nearby, and radio delay makes ground control impractical. Such a machine must run itself, which means it needs a capable onboard computer. Onboard computers in space, however, are hit by radiation, which flips bits in memory and logic and slowly degrades the device.

These two problems are studied separately. Researchers who build self-correcting 3D printers assume the computer running the printer never fails [3, 4]. Researchers who protect processors against radiation test them on standard benchmark programs such as CoreMark [5–7]. **Nobody has studied the overlap: what happens to a self-correcting printer when radiation corrupts the computer controlling it, and whether a protected processor can still meet its timing deadlines while running a machine-learning inspection step.**

This project sets out to build and measure that overlap. I will take an open-source RISC-V processor and add protection against radiation faults, using error-correcting memory, redundant copies of critical logic, two cores running in step so they can check each other, and watchdog timers. That processor will then be the only computer controlling a 3D printer testbed built to behave the way one would in microgravity, with cameras and thermal sensors closing the loop. Finally, I will deliberately inject faults — both radiation-like faults in the computer and physical faults in the printer — and measure what happens.

The work is scoped to twelve months, reaches Technology Readiness Level 3–4, and needs about Rs. 67,200 of equipment. It is an FPGA design and verification project at heart, which places it squarely within this group's existing work on FPGA-based signal, image and cryptographic hardware, and it opens up space electronics as a new direction for the Department.

2 Background and Motivation

2.1 The case for manufacturing in orbit

Everything in orbit had to be launched there, and launch is expensive. Spare parts and tools are carried up as dead weight for failures that may never happen, and any structure has to fit inside a rocket's nose cone. Manufacturing in orbit changes this. Instead of many specific spare parts, you carry one generic material; instead of a warehouse, you send design files; and structures no longer have to fit inside a nose cone [8].

This is not a speculative idea. NASA and Made In Space first printed plastic parts on the ISS in 2014 [2]. A permanent printer, the Additive Manufacturing Facility, was installed in 2016 and has since made more than 200 parts in orbit [1]. The European Space Agency printed the first metal part in space in 2024 [9]. The United States published a national strategy for in-space manufacturing in 2022 [10], and NASA's additive manufacturing standard, NASA-STD-6030, now covers printing in space directly [11].

The manufacturing technology is mature. The autonomy is not.

2.2 Autonomy as the limiting capability

Today's space printers always have a person involved. Print jobs are sent up from the ground, the crew keeps an eye on them, and someone fixes the machine by hand when a print goes wrong. The things that go wrong are ordinary and frequent: the filament jams or slips, the part warps or lifts off the bed, a layer shifts, the nozzle clogs. On Earth, anyone watching the printer can fix any of these in a minute.

Take the person away and every one of them can end the mission. A printer on the Moon or on a satellite has to spot its own defects, work out what caused them, fix them, and decide when

a print is not worth continuing — on its own, for weeks at a time. That takes real computing power on board: precise timing control of the motors and heaters, combining data from cameras, thermal sensors and accelerometers, a machine-learning model to recognise defects, and a layer of logic above all of it to detect faults, work out where they came from, and recover from them.

2.3 Why onboard computation is difficult in the space environment

Processors approved for space are usually a generation behind ordinary chips, because making them radiation-tolerant is slow and expensive. Charged particles striking the chip flip individual bits (a single-event upset), cause brief glitches in logic (a transient), lock up a whole subsystem (a functional interrupt), or trigger a damaging short circuit (latch-up). Separately, the total radiation dose absorbed over the mission gradually degrades the device.

RISC-V has changed the cost of dealing with this. Because the instruction set is open, protection can be built into the processor itself instead of being bolted on around a chip whose internals nobody outside the vendor can see [7, 12]. NASA and Microchip are building exactly such a chip, the High Performance Spaceflight Computer, a fault-tolerant multi-core RISC-V design now in radiation, thermal and shock testing [13], and Frontgrade Gaisler's NOEL-V is a European equivalent already in use [12]. University groups have followed. Protected RISC-V designs have been tested in neutron and proton beams, and the standard technique of running three copies of the logic and voting on the result — triple modular redundancy, or TMR — has been measured to cut the error rate by about 33×. It costs roughly 27 % of the clock speed and 5.6× the logic area, and the processor gets about 24× more work done before it fails [5].

India has a direct stake in this. For two decades ISRO has used the 15UD Pelletron accelerator at the Inter-University Accelerator Centre (IUAC) in New Delhi to test FPGAs, memories and custom chips for radiation effects before flying them, on missions including Chandrayaan and the Mars Orbiter Mission [14]. The ability to design fault-tolerant processors at home matters strategically, and at present only a few institutions in the country work on it.

3 Process Selection: What Works in Microgravity, and Why

A 3D printer in a university lab is not a space manufacturing system, and the gap is not just a matter of paperwork. This section works out which printing processes still function in free fall, explains why, and lists what has to change in the machine itself. It comes early because it decides how much mechanical work the project involves.

3.1 Why material extrusion is gravity-insensitive

In fused filament fabrication (FFF), a motor pushes solid filament into a hot nozzle, and it is that mechanical push — not gravity — that forces the molten plastic out. The shape the molten bead takes as it lands is set by a contest between gravity, which pulls it down, and surface tension, which holds it together. The ratio of the two is the Bond number:

$$\mathrm{Bo} \;=\; \frac{\rho g L^2}{\sigma}, \tag{1}$$

Here ρ is the density of the molten plastic, g is gravity, L is the size of the bead and σ is its surface tension. A Bond number well below one means surface tension wins and gravity barely matters. For a typical printed bead ($\rho \approx 1000\,\mathrm{kg/m^3}$, $L \approx 0.4\,\mathrm{mm}$, $\sigma \approx 0.03\,\mathrm{N/m}$) the answer is $\mathrm{Bo} \approx 0.05$ — and that is *on Earth*. Surface tension already beats gravity by more than a factor of twenty at the scale that decides bead shape. The plastic is also extremely viscous, around $10^2\,\mathrm{Pa\,s}$ to $10^3\,\mathrm{Pa\,s}$, which resists gravity-driven flow even further in the fraction of a second before the bead solidifies. Switching gravity off therefore removes a term that was already close to irrelevant.

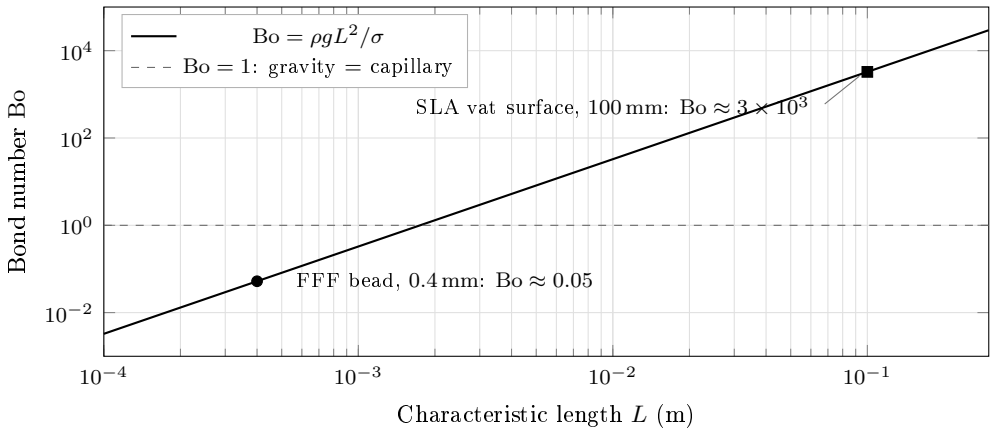


Figure 1: How much gravity matters, as a function of size, for molten plastic ($\rho = 1000\,\mathrm{kg/m^3}$, $\sigma = 0.03\,\mathrm{N/m}$). Below the dashed line, surface tension wins and gravity is negligible. A printed bead sits well below it even on Earth, which is why extrusion still works in orbit. The open surface of a resin tank sits far above it, which is why resin printers do not.

Experiment supports this reasoning. NASA compared parts printed on the ISS with identical parts printed on the ground and did find differences in density and strength. When they investigated, the cause turned out to be a small difference in machine settings — the gap between the nozzle and the print bed — and not microgravity affecting the printing process [2]. **This result is what makes the present proposal possible: it means the process can be developed on the ground, so long as machine settings are controlled and recorded just as carefully.**

3.2 Why the alternative processes do not transfer

The question here is not which printing process makes the best parts, but which ones still work at all once gravity is gone. Table 1 is therefore arranged by what breaks, not by part quality. In every case that fails, the reason is the same: the process uses gravity to *put material where it belongs* — to settle a flat layer of powder, to hold a level pool of liquid resin, to drop a droplet onto a target. No amount of tuning replaces that. Extrusion is the exception, because it places material mechanically and holds it there with surface tension and viscosity, neither of which cares about gravity.

Material extrusion is therefore selected: it is the only family that is simultaneously flight-proven, free of loose particulate, modest in power demand, and benign in its failure modes [15,16].

3.3 What must nevertheless change in the machine

The fact that the *printing physics* survives free fall does not make a lab printer a space printer. The Department's existing machines cannot be modified, and in any case a commercial printer is the wrong starting point: its frame, airflow and control electronics are all designed around assumptions that this project needs to break. The machine will therefore be designed and built from components. That is more work, but it buys the freedom to put sensors where they are needed, to seal and instrument the chamber properly, and — most importantly — to let the SoC drive the motion hardware directly rather than through a closed commercial controller. The following changes are what the custom build is for, and they constitute the mechanical work of this project.

1. **Cooling without air.** On Earth, hot air rises and carries heat away on its own. In free fall it does not rise at all, so that mechanism disappears; and on an unpressurised platform there is no air to move in the first place. Heat must then leave by conduction and radiation alone. How fast a bead cools decides how well it bonds to the layer below and how much stress is locked

Table 1: Which 3D printing processes still work without gravity.

Process family	**Viability**	**Governing reason**
Material extrusion (FFF)	Demonstrated	Melt confined and mechanically driven; capillary-dominated bead formation; adhesion by thermal fusion. Flight heritage since 2014 [1, 2].
Powder bed fusion (SLS, L-PBF)	Not viable near-term	Requires a gravity-settled powder layer spread by a recoater. Loose powder disperses in free fall and constitutes an inhalation, contamination and electrical hazard in a pressurised volume.
Vat photopolymerisation (SLA)	Conditionally viable	Requires a flat free liquid surface, which does not form in microgravity; demands a constrained or membrane-bottomed vat and resin containment.
Binder jetting	Not viable near-term	Combines the powder-bed problem with unassisted droplet placement.
Directed energy deposition, wire-fed	Demonstrated, high cost	Wire is mechanically fed and the melt pool retained by surface tension; ESA demonstrated metal printing aboard the ISS in 2024 [9]. High power, high thermal load and spatter containment place it beyond an undergraduate project.
Bioprinting	Benefits from microgravity	Absence of gravitational collapse removes the need for scaffolding; outside the scope of this proposal.

into the part, so this is not a detail — it changes the thermal design of the whole machine. Section 3.4 sets out how it is handled here.

2. **Containing what the plastic gives off.** Melting plastic releases fumes and very fine particles. In a pressurised cabin that is a crew-health problem; in vacuum it becomes a molecular contamination problem, which is why outgassing is a standard space-materials screening requirement. Either way the chamber must be closed and the emissions captured or pumped away.
3. **Everything must be held down deliberately.** Nothing stays put under its own weight. The part must be gripped to the bed, released in a controlled way, and caught once released, and any debris must be captured — otherwise it simply floats away.
4. **Motors have to sit outside the chamber.** This one bites immediately. Stepper motors are cooled almost entirely by convection into the surrounding air; remove it and heat leaves only by conduction through the mounting face, so continuous torque must be derated substantially — suppliers of vacuum-rated motors quote reductions of 20 % to 50 % and sometimes a factor of several, depending on duty cycle and mounting. Vacuum-rated motors exist but are expensive. The design instead keeps the motors outside the vacuum boundary and drives the mechanism through the wall. CoreXY makes this natural: both X and Y motors are frame-mounted and stationary by construction, so only belts cross the boundary. It does, however, argue for a Bowden extruder rather than direct drive, since direct drive would put a motor on the moving carriage inside the chamber.
5. **A controlled filament path.** On Earth the spool stays neat and the filament stays taut partly because of gravity. In orbit it will not, so the feed needs active tension control and a sensor to detect the filament slipping.
6. **It has to survive the launch.** The machine will be shaken far harder on the way up than any lab instrument ever is, which changes how stiff the frame must be and how everything is fastened.

3.4 Thermal control without air

Blowing air across the part is how every desktop printer manages cooling, and it is the wrong answer here. It is worth being precise about why, because fans are not universally wrong in space:

inside a pressurised module there is cabin air, forced convection works, and it is the standard flight-proven solution — the printers currently aboard the ISS rely on it. But this project's stated target is the case where crew are absent: a free-flying platform, an external payload, or a lunar surface asset. There, either there is no air at all, or the volume is small and sealed and a fan is simply recirculating its own contamination. A fan also has bearings, which is a wear item in a machine whose entire premise is unattended operation for weeks, and it injects broadband vibration and airflow noise into exactly the accelerometer and acoustic channels this project uses to detect nozzle clogs.

The machine is therefore designed around conduction and radiation, and the chamber is built to be evacuated rather than ventilated.

Radiation is not negligible at extrusion temperatures. It is easy to dismiss radiative cooling as a small effect, but the fourth-power dependence rescues it. For a freshly deposited bead at 240 °C (513 K) radiating to chamber walls held at 60 °C, with emissivity $\varepsilon \approx 0.9$,

$$q'' = \varepsilon\sigma\left(T_{\text{bead}}^4 - T_{\text{wall}}^4\right) \approx 0.9 \times 5.67 \times 10^{-8}\left(513^4 - 333^4\right) \approx 2.9\,\text{kW/m}^2. \tag{2}$$

That is a real heat flux, and it is controllable: chamber wall temperature and wall emissivity become design variables, and the same equation says that a 20 °C change in wall temperature is a usable control input.

The thermal architecture.

- **Hot end:** heat is conducted off the heat break rather than blown off it. The published vacuum-FFF systems use a pumped liquid cold end [17, 18], and that remains the fallback, but the preferred solution here is a sintered-wick heat pipe. Heat pipes are capillary-driven, contain no pump, and are entirely gravity-independent — which is why they are standard on spacecraft, and why they suit a machine intended to run unattended for weeks. Removing the pump removes the last rotating part in the thermal path.
- **Part cooling:** conduction down through the deposited layers into a temperature-controlled build plate, plus radiation to the chamber walls. There is no impingement cooling on the bead.
- **Chamber walls:** held at a commanded temperature by resistive film heaters under closed-loop control. Thermoelectric modules were considered and rejected: for ABS and polycarbonate the chamber wants to be *hot* — around 60 °C, to stop the part warping — so the cooling direction would essentially never be used, and a thermoelectric stage would draw over 100 W to do what a film heater does for a fraction of it. Heat leaves the chamber passively to the laboratory through the wall.
- **Layer time as the primary cooling control.** This is the consequence that matters for the control problem. Without convection, the fastest available cooling mechanism is gone, so interlayer temperature is regulated by *how long the machine waits* — dwell time, layer scheduling, and toolpath ordering — rather than by how hard a fan blows. Thermal management moves from the actuator domain into the planner, which makes it a scheduling problem for the controller running on the SoC.

How much vacuum is actually required. It is worth deriving this rather than assuming, because the answer sets the single largest item in the budget. Buoyancy-driven convection is governed by the Rayleigh number,

$$\text{Ra} = \frac{\rho^2 g \beta\, \Delta T\, L^3 c_p}{\mu k}, \tag{3}$$

and the relevant feature is the ρ^2 dependence: halving the gas density quarters the tendency to convect. For this chamber ($L \approx 0.15$ m, $\Delta T \approx 180$ K between a 240 °C bead and 60 °C walls)

the Rayleigh number at atmospheric pressure is of order 10^7. Convective cells do not form below $\mathrm{Ra} \approx 10^3$, so a reduction of about four orders of magnitude is required, which corresponds to a hundredfold fall in density — roughly 10 mbar.

The chamber is therefore specified at 1 mbar, which carries an order of magnitude of margin over the requirement.

One limitation should be stated rather than glossed over. Suppressing *convection* is not the same as removing the gas. At 1 mbar the mean free path in air is around 65 µm, still short compared with millimetre-scale gaps, so residual gas *conduction* persists; it falls away only below roughly 0.1 mbar. The chamber therefore reproduces the absence of buoyancy-driven convection exactly — which is the microgravity condition, and is what the ISS printers use fans to compensate for — while only partially reproducing the conduction environment of an unpressurised platform. The pressure sweep of Section 3.5 maps both regimes, and the distinction is recorded here rather than discovered in review. This matters practically: 1 mbar is reached by a single-stage rotary-vane pump costing a fraction of the two-stage pump needed for 10^{-3} mbar, and it needs only a simple analogue gauge rather than a Pirani head. The published vacuum-FFF work operates far lower — down to 10^{-4} mbar [17] — but those experiments are also characterising outgassing and contamination, which this project is not. Specifying more vacuum than the physics requires would consume roughly a fifth of the total budget for no experimental gain.

Borrowed directly from spacecraft thermal practice. None of this needs inventing. Satellites have solved the no-convection problem for sixty years, and the following are cheap, available, and directly applicable:

- **Controlled surface emissivity.** Emissivity appears explicitly in the radiative flux above, so it is a design variable rather than a material accident. High-emissivity coating on the chamber interior maximises heat absorbed from the part; polished surfaces isolate what should stay hot. It also becomes a cheap experimental variable: sweeping wall emissivity and measuring the resulting cooling rate is a direct check on the ANSYS radiation model.
- **Flexible thermal straps.** Braided copper is the standard spacecraft answer for conducting heat across a joint that has to move.
- **Multi-layer insulation.** Aluminised film around the hot end and chamber exterior. Worth noting that MLI only works properly *in vacuum* — in air the trapped gas conducts straight through it — so this is one of the few benchtop machines where it earns its place.
- **Isolation standoffs.** Low-conductivity mounts (G10 or PEEK) to stop heat leaking into the frame, where it would distort the structure and detune the sensors.
- **Heaters, not just cooling.** Most spacecraft thermal control is about keeping things warm. The same applies here: with convection gone and walls held cold for part cooling, the chamber will need trim heating to hold the filament path and sensors in range.

The problem this creates: the hot end moves. On Earth this is invisible, because the cooling fan simply rides on the carriage. Conducting heat off a moving gantry is harder, and there are only three ways to do it — braid, a heat pipe with a compliant section, or fluid tubing with a service loop. Each trades thermal conductance against the drag force it adds to the motion system, and that drag directly affects positioning accuracy and therefore the quality of the layer imagery the whole inspection loop depends on. This is a real trade rather than a detail, and it is resolved as an explicit task in WP3 (Table 7).

This is established, not speculative. FFF has been demonstrated at 10^{-4} mbar, where convective heat transfer is negligible, using active water cooling to keep components from overheating [17]. Polycarbonate has been printed under low vacuum with no loss of mechanical performance and with contamination low enough to characterise [19]. High-temperature space-relevant

polymers — PEKK, PEI and their carbon-nanotube composites — have been printed in high vacuum [20], and for PEEK, layer adhesion actually *improves* as pressure falls [18]. That last group also report that low-cost vacuum equipment is sufficient, which is what makes this approach viable on an undergraduate budget.

One finding worth flagging in advance. The vacuum FFF work reports that the hot-end PID parameters had to be retuned substantially for vacuum operation, because the heat-loss path changes completely [17]. That is a control-design problem, it lands squarely in the MATLAB work of Section 7.2, and it is a small original contribution available cheaply: nobody has characterised how a *closed-loop, sensor-driven* thermal controller must change when convection is removed.

3.5 Establishing microgravity representativeness without flight access

Gravity cannot be switched off on a bench, but its two consequences can be attacked separately, and the evacuable chamber does most of the work.

Removing convection is not an approximation — it is exact. As shown in Section 3.4, the Rayleigh number falls as ρ^2, so a chamber held at 1 mbar suppresses convection with an order of magnitude of margin. A ground chamber at low pressure therefore reproduces the *thermal* environment of free fall faithfully rather than approximating it, and reproduces an unpressurised orbital platform exactly. Chamber pressure becomes an experimental variable in its own right: printing the same specimen at a sequence of pressures from ambient down to 1 mbar maps where convection ceases to matter and gives a direct experimental check on the ANSYS zero-gravity thermal model rather than leaving it unvalidated. The predicted transition near 10 mbar is itself a falsifiable claim, which is better than an assumption.

Removing the body force on the melt is the part that cannot be reproduced on the ground, so it is addressed by argument and by *orientation testing*: print the same specimen with the machine on its side and inverted. If results do not change when gravity points elsewhere, they are unlikely to change when it is absent — and Section 3.1 already shows why this should be expected at the bead scale. Alongside this, all comparison specimens are printed from one filament batch with identical recorded settings, following NASA's method [2].

Between them these cover the two mechanisms, and the pressure sweep is the stronger of the two because it is a measurement rather than an inference. It is also worth being clear about what the chamber is: a *simulator*, not a prototype subsystem. Its purpose is to reproduce on a bench the pressure that an unpressurised orbital platform provides at no cost, so the pressure sweep is a ground-only diagnostic that validates the thermal model. A flight unit would simply operate at the low end of that sweep permanently (Section 8). A parabolic flight would still add value, but it is follow-on work and nothing here depends on it.

4 Research Gap

I surveyed three separate research fields. Each one is well developed on its own, but the place where they meet has not been studied (Table 2). Each field, it turns out, assumes away exactly the problem that one of the others works on.

Gap 1: the test programs are not realistic. The cost of protecting a processor is always measured while it runs a benchmark program. A printer controller behaves nothing like a benchmark. It has to respond within fixed deadlines, it reads sensors continuously, and it runs a neural network in the middle of the control loop. Losing 27 % of the clock speed barely matters to a benchmark, which simply finishes later. For a motion controller, the same slowdown may mean a deadline is missed and the machine misbehaves. Nobody has measured the cost of protection while running a real manufacturing control workload.

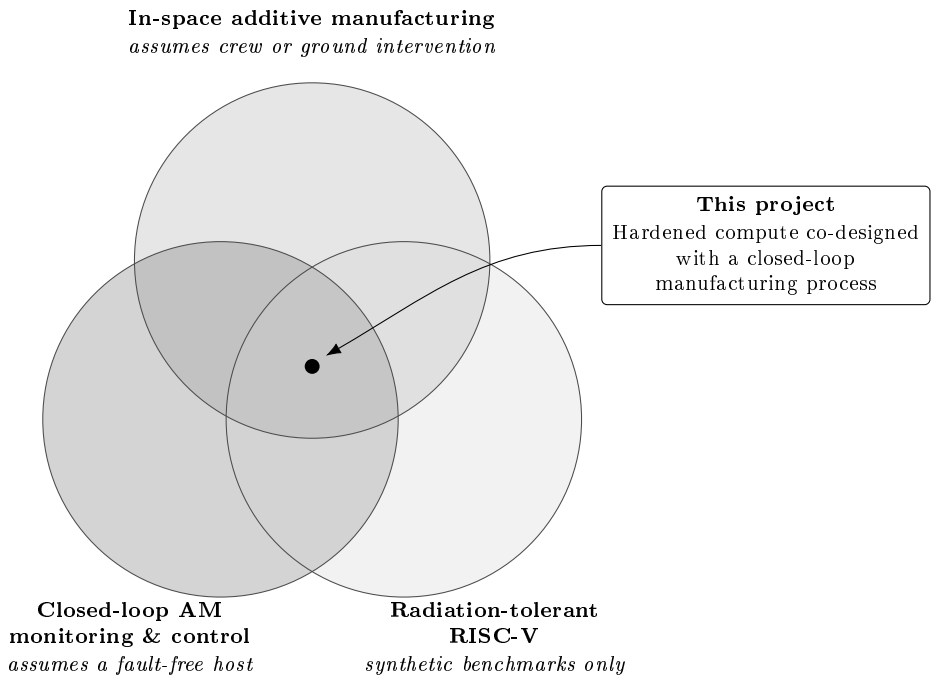


Figure 2: The three fields this project draws on, and the gap where they meet. Each field is mature on its own, and each takes for granted the very thing the next one studies.

Table 2: Maturity and assumptions of the three contributing literatures.

Literature	**Mature in**	**Assumes away**
In-space additive manufacturing	Process feasibility in microgravity; material characterisation; flight heritage [2, 15, 16]	Autonomy: presumes crew or ground intervention
Closed-loop AM monitoring and control	Optical, thermal, acoustic and vibration sensing; CNN and SVM defect classifiers reporting 90 % to 95 % accuracy; autonomous parameter correction [3, 4, 21, 22]	Computing platform reliability: presumes a desktop-class, fault-free host
Radiation-tolerant RISC-V	TMR, ECC, lockstep, scrubbing, fault awareness; beam-validated cross-sections [5, 6, 23, 24]	Workload realism: benchmarked on CoreMark, Embench and Dhrystone; no real-time control or ML inference under fault

Gap 2: the inspection step can fail silently. A self-correcting printer acts on whatever its defect classifier reports. If radiation flips a bit inside that neural network — in one of its weights, in an intermediate value, or in the threshold it compares against — the program does not crash and no error is reported. It simply gives the wrong answer, with full confidence. The printer then "fixes" a defect that was never there, or ignores a real one. Error-correcting memory protects stored data, but nobody has measured how often a bit flip changes the final decision a printer acts on, or tested what kind of redundancy protects that particular step.

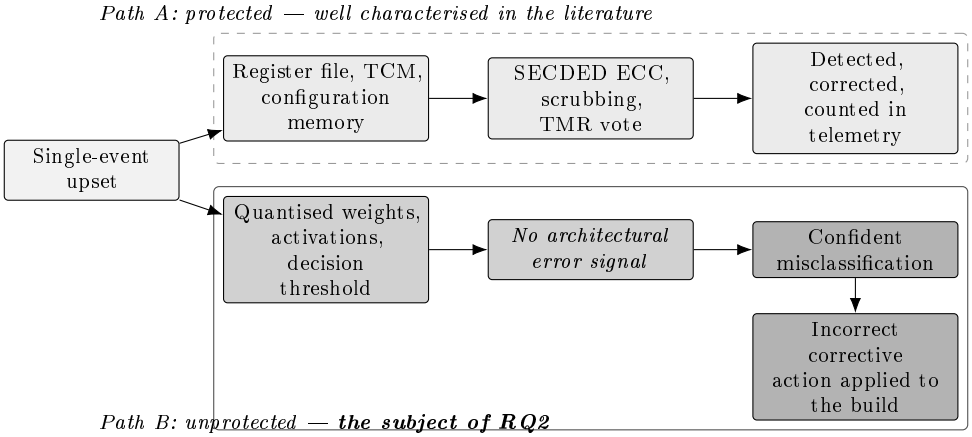


Figure 3: What happens when radiation flips a bit inside the printer's controller. If the bit is in memory or control logic (Path A), existing protection catches it, fixes it, and records it. If it is inside the neural network (Path B), nothing catches it: the program does not crash, it just returns a wrong answer confidently, and the printer then acts on that wrong answer. How often this happens has never been measured.

Gap 3: nothing tells the two kinds of fault apart. Faults in the computer and faults in the printer are handled by completely separate machinery, and neither knows about the other. The trouble is that they look identical from outside. A jammed extruder and a crashed control program both show up as "no plastic is coming out". If the machine guesses wrong about which one it is, it applies the wrong fix — resetting the processor will not clear a jam. No published design reasons about both kinds of fault together.

These three questions are genuinely open, and answering them needs about Rs. 67,200 of equipment. Problems that are both unsolved and cheap to attack are rare, which is why I am proposing this rather than a more conventional VLSI topic.

5 Research Questions and Hypotheses

RQ1. Which combination of protection techniques gives the most reliability per unit of chip area and speed, for a small RISC-V processor running a real 3D printing control workload?
H1. Protecting only the parts that matter most — triplicating the control logic and status registers, using error-correcting memory, and running two cores in step to check each other, while leaving the arithmetic unit unprotected — will get within 20 % of the reliability of triplicating everything, at less than half the cost in chip area, and will still meet the timing deadlines that full triplication would break.

RQ2. When radiation flips a bit inside the neural network that inspects each layer, what happens, and how often does the printer end up making the wrong decision as a result?
H2. A significant share of bit flips in the neural network will produce a wrong answer with no error reported at all, and that share will be larger than the share that produces

a visible crash. If so, the inspection step will turn out to cause more bad decisions than any other part of the system, even though it is the smallest part of it.

RQ3. Can one system reliably tell a computer fault from a printer fault and apply the right fix to each, and how accurately?

H3. Combining two sources of evidence — the error counters inside the processor and the physical sensor readings from the printer — will tell the two kinds of fault apart far better than either source can on its own.

RQ1 and RQ2 are what I commit to answering in twelve months. RQ3 will be attempted if there is time; if not, it is the obvious next step.

6 Objectives

The objectives below turn the research questions of Section 5 into concrete things that can be built and checked. Each one has a clear pass or fail test, so progress at the review meetings in Section 8 can be judged on evidence rather than opinion.

The **Core** objectives are what I commit to in twelve months, and together they answer RQ1 and RQ2. The **Extension** objective answers RQ3 and will only be attempted if the core work finishes early. I have listed it rather than leaving it out, because you should be able to see where the project leads before agreeing to its first year.

O1 and O2 are the hardware; O3 and O4 are the printer and its control loop; O5 and O6 are the experiment itself. O1 is checked at month 2 (Section 7.3), and that is the one point at which the project should be stopped if it is not working.

Table 3: Project objectives and success criteria.

#	**Objective**	**Success criterion**	**Tier**
O1	Establish a verified RISC-V SoC baseline on FPGA	Boots an RTOS; passes the `riscv-arch-test` compliance suite via RISCOF; continuous-integration pipeline operational; bus adapter defined so that the core remains replaceable	Core
O2	Implement a parameterised hardening architecture (ECC, selective TMR, DCLS, scrubbing, watchdog escalation, error counters)	All mitigations independently enabled or disabled at build time, permitting controlled comparison	Core
O3	Design, analyse and build a microgravity-representative FFF machine	CAD model and ANSYS structural and thermal analysis complete; machine built and printing repeatably; evacuable chamber with conduction and radiation thermal control; optical, thermal, vibration and feed sensing	Core
O4	Implement closed-loop defect detection and autonomous correction	$\geq$ 4 defect classes detected; $\geq$ 90 % classification accuracy after quantisation; corrective actions bounded by a hardware safety envelope	Core
O5	Develop an open fault-injection framework	Injects SEU and multiple-bit upsets into RTL simulation and FPGA configuration and user memory; injects physical process faults reproducibly	Core
O6	Execute the characterisation campaign; answer RQ1 and RQ2	Statistically adequate sample sizes with reported confidence intervals	Core
O7	Address RQ3: unified compute/process fault discrimination	Discrimination accuracy quantified against single-modality baselines	Extension
O8	Disseminate	Two submitted papers; open-source RTL, framework and dataset	Core

7 Methodology

7.1 Overall approach

The hardware will be developed using the European Space Agency's standard process for FPGA and chip design, ECSS-E-ST-20-40C, together with its quality companion ECSS-Q-ST-60-03C [25, 26]. The choice of protection techniques follows the ESA handbook on radiation effects [27]. Measurements follow ordinary controlled-experiment practice: change one thing at a time and compare against a fixed baseline.

Following a recognised space standard from day one is deliberate. It costs very little extra now, while the project is small. Adding it later, once the design already exists, is painful — and without it, a design can never be put forward for a real flight, however good it is.

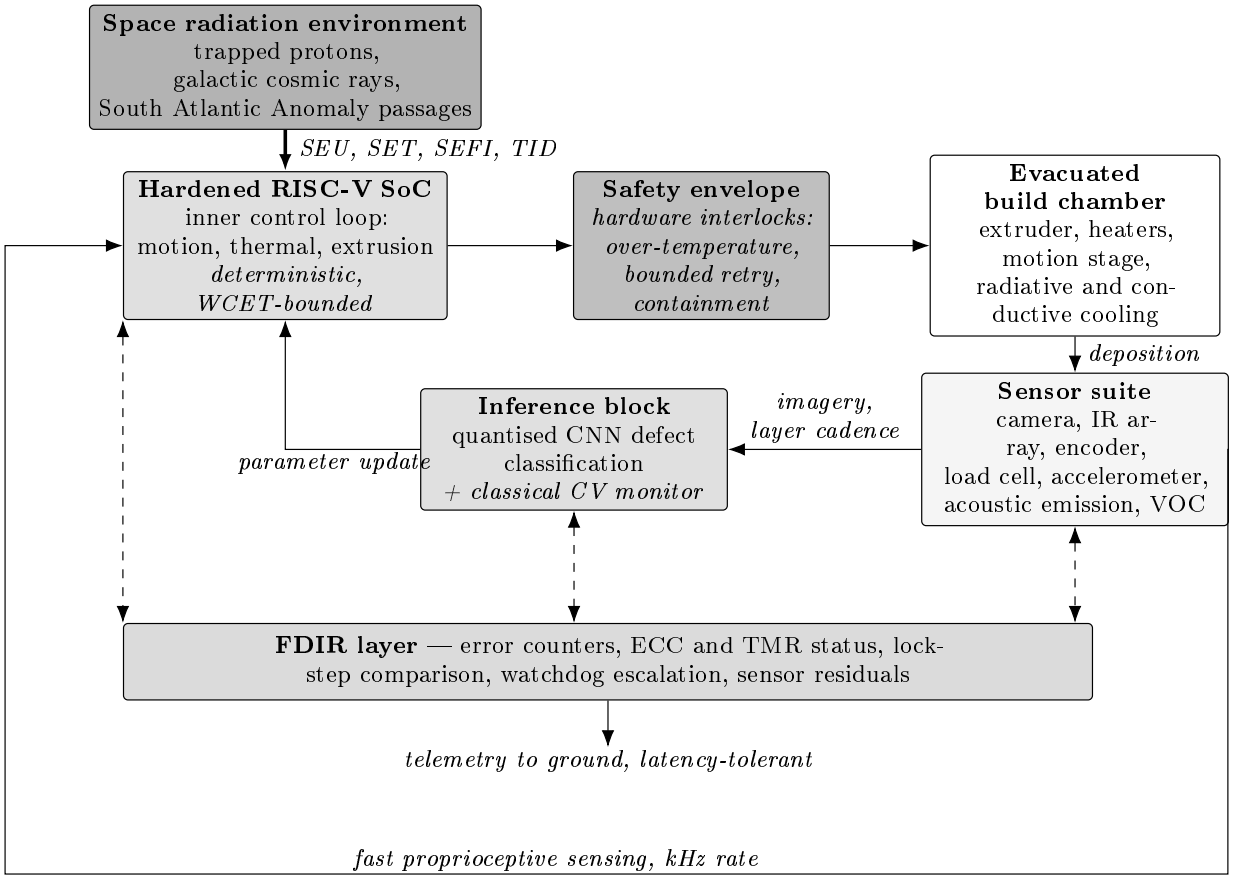


Figure 4: End-to-end system pipeline. Radiation acts on the controller, not on the process: charged particles upset the SoC, whose corrupted output then propagates forward through the safety envelope into the physical build, and returns through the sensor suite. Two return paths operate at different rates — fast proprioceptive sensing closes the inner control loop, while imagery closes the slower inspection loop through the inference block. The FDIR layer observes all three stages, which is what makes fault-origin discrimination (RQ3) possible at all.

7.2 Design and analysis tools

The work spans four domains, and each is handled with the standard tool for it rather than with improvised alternatives.

- **SolidWorks — mechanical design.** The full printer, chamber, filament path and sensor mounts are modelled in CAD before anything is cut. This produces the fabrication drawings, the bill of materials, and the geometry that feeds directly into the analysis below.

- **ANSYS — structural and thermal analysis.** Two uses. Structural: modal and random-vibration analysis of the frame and chamber, so that the design can be assessed against launch-representative loads by analysis even though no shaker test is planned within the twelve months. *Thermal: this is where the microgravity argument becomes a design calculation rather than a claim.* The chamber is modelled in three configurations — at $1\,g$ and atmospheric pressure, with the gravitational body force set to zero, and at reduced pressure — to show that the evacuated ground chamber reproduces the free-fall thermal environment. Conduction and surface-to-surface radiation are the transport mechanisms; there is no convection term to size a fan against. What the analysis produces instead is the build-plate power, the chamber wall temperature setpoint, the required wall emissivity, the minimum layer dwell time needed to reach the target interlayer temperature, and the conductance required of the moving thermal link (Section 3.4).
- **MATLAB and Simulink — control design.** Plant identification from measured step responses, controller design and discretisation for the motion, thermal and extrusion loops, stability margins, and hardware-in-the-loop testing of the control law before it is committed to the SoC. Because the heat-loss path changes completely once convection is removed, the thermal loops are identified and tuned separately at atmospheric pressure and under vacuum — published vacuum-FFF work reports that hot-end PID parameters differ substantially between the two [17]. The Simulink plant model doubles as the reference against which sensor residuals are computed, which is what makes the process-fault branch of Figure 8 possible.
- **FPGA and verification toolchain.** AMD Vivado for the SRAM-based device and Microchip Libero for the flash-based device; Verilator for fast simulation; RISCOF and `riscv-arch-test` for instruction-set compliance; `riscv-dv` and Spike for randomised co-simulation. All of the verification tooling is open-source.

Vivado and Libero are free at the editions required. SolidWorks, ANSYS and MATLAB are already licensed to the Institute for teaching use, so no new software purchase is requested.

7.3 WP0 — Planning (months 1–2)

I am entering the third year of the B.E. programme and have not yet written hardware description code at any serious scale. Rather than gloss over that, the first two months are set aside openly as a learning phase, with specific things to deliver and a real stop/continue decision at the end of month 2.

- **Weeks 1–4: learn the tools.** SystemVerilog, plus the Verilator and Vivado (or Libero) flow. To prove it works, I will build and test two small blocks — an arithmetic unit and a serial port — with automatic tests, and run them on a real board.
- **Weeks 5–8: get a RISC-V processor running.** Put the chosen open-source core onto the board, run the official RISC-V test suite against it, boot a small real-time operating system, and set up automatic testing so every change is checked.
- **Throughout: finish the literature review.** Search IEEE Xplore, Scopus, ScienceDirect and NASA's technical reports using a search plan fixed in advance, and expand Table 2 into a full survey.

The month 2 decision. If the processor is not running the test suite on real hardware by the end of month 2, the project should be cut back or stopped then — not at month 10, by which point far more time has been spent. I would rather this be applied strictly than generously.

7.4 WP1 — Build the SoC (months 2–5)

Base core: Ibex. I will not write a processor from scratch. Writing one teaches you a great deal, but it is not a research result — hundreds already exist, and mine would be worse. The new

work here is the protection scheme and the measurements, so the project starts from a proven open-source core.

The core is lowRISC's Ibex [28], a small 32-bit RISC-V CPU used in OpenTitan. The reason is specific rather than general: Ibex already implements, as configuration options, most of the protection mechanisms this project sets out to measure. Turning them on is a parameter; *measuring what they cost, and what they miss, is the research.* Building them myself would consume the schedule and produce implementations a reviewer would reasonably distrust.

Table 4: Base core comparison. Ibex is selected; the others remain named fallbacks.

Core	Language	Assessment
Ibex (lowRISC)	SystemVerilog	**Selected.** Dual-core lockstep, register-file and cache ECC, hardened PC, shadow CSRs and alert outputs all exist as parameters (Table 5). Strong CI, riscv-dv and formal verification behind it. Deployed in OpenTitan, so the implementations are scrutinised.
NEORV32 [29]	VHDL	*Fallback.* Complete SoC rather than a bare core, so bring-up is faster — but every protection mechanism would have to be written and then defended. Reachable in days via the bus adapter (Figure 5).
CV32E40S (OpenHW)	SystemVerilog	*Fallback.* Safety and security variant with an industrial UVM environment, but `core-v-verif` is a large thing to learn alongside everything else.
CVA6 [30]	SystemVerilog	*Excluded.* Application-class; expects external DRAM, which the Boolean Board does not have.

What Ibex already provides. Setting the `SecureIbex` parameter enables a set of features developed for security-critical use, which map almost directly onto radiation-tolerance needs [31]. Table 5 lists them alongside the fault class each addresses.

What this leaves to do, and why it is still a project. Two things. First, **Ibex is a core, not a system.** It has no bus, no memory controller, no peripherals. The SoC around it — bus fabric, block-RAM and QSPI controllers, UART, timers, the sensor interfaces, the inference block — is mine to build, and that is where the protection mechanisms in the lower half of Table 5 live. The reference integrations (`examples/simple_system` and the Ibex Demo System) provide a working skeleton to start from rather than a blank sheet.

Second, and more importantly, **Ibex's features detect faults; they do not tell you what a fault means to a manufacturing process.** A major alert says the lockstep pair diverged. It does not say whether the print is recoverable, whether the last layer should be reprinted, or — the RQ2 case — whether a classifier verdict that raised *no* alert should nonetheless be distrusted. Ibex hands this project a well-built signalling layer. The contribution is what is built on top of it.

Relevant prior art. This approach is not unprecedented, and the closest existing work is worth naming explicitly. A group at CERN asked whether Ibex's built-in security features can be used to detect single-event upsets, using a cocotb testbench against the Ibex RTL [32]. That work validates the premise — security hardening transfers to radiation tolerance — and stops where this project begins: it characterises detection *within the core*, on a synthetic workload, with no control loop, no physical process, and no machine-learning path. Gaps 1 and 2 of Section 4 survive it intact.

Configuration on the Boolean Board. Ibex will be built as RV32IMC with `RegFile = RegFileFPGA`, the implementation intended for FPGA targets rather than the flip-flop or latch variants used for ASICs. Ibex is small — a few thousand LUTs unprotected — so even with

Table 5: Ibex security features and the radiation fault classes they address. The rightmost column is the honest accounting of what is adopted versus what this project contributes.

Mechanism	**What it catches**	**Origin**
Dual-core lockstep	Shadow core runs on delayed inputs; outputs compared against delayed main-core outputs; mismatch raises a major alert. The delay *is* temporal diversity, so common-mode transients are caught too	Ibex
Register-file ECC and write-enable glitch detection	Upsets in architectural state	Ibex
One-hot register addressing with checkers	Glitches in read-address decode	Ibex
Instruction-cache ECC	Upsets in cached instructions; raises a minor alert	Ibex
Hardened PC	Checks that the IF-stage PC equals the previous PC plus the correct increment — catches control-flow corruption	Ibex
Shadow CSRs	Complemented duplicate of critical CSRs with continuous consistency checking	Ibex
Bus integrity checking	Corruption on the memory interfaces	Ibex
Minor and major alert outputs	The signalling layer the FDIR logic subscribes to	Ibex
Memory scrubbing	Accumulated upsets in idle block RAM	**This project**
TMR on peripheral control FSMs	Upsets outside the core, which Ibex does not cover	**This project**
Staged watchdog escalation	Task → application → core → system reset	**This project**
Error counters as telemetry	Turns alerts into a time series the FDIR layer and RQ3 can reason over	**This project**
Redundancy on the inference path	The RQ2 gap: Ibex protects the processor, not the meaning of what it computes	**This project**

lockstep doubling the core logic, the XC7S50's 32,600 LUTs leave substantial room for the SoC and the inference block. This is a more comfortable resource position than the full-TMR estimate of Section 7.5 implied, and it is another reason the choice is Ibex.

Insulating the project from the core choice. Selecting Ibex over NEORV32 buys mature protection mechanisms at the cost of a harder start: Ibex is a bare core, whereas NEORV32 arrives as a complete SoC. That trade is worth making, but it concentrates schedule risk in the first two months, and the standard mitigation — "keep NEORV32 as a fallback" — is weak if switching means rebuilding everything.

The design therefore places a **thin bus adapter** between the core and everything I write. All project-built modules — sensor interfaces, the QSPI weight streamer, the block-RAM scrubber, the watchdog and error counters, the inference block — target a single internal bus definition rather than Ibex's native interface. The adapter is the only module that knows which core is underneath.

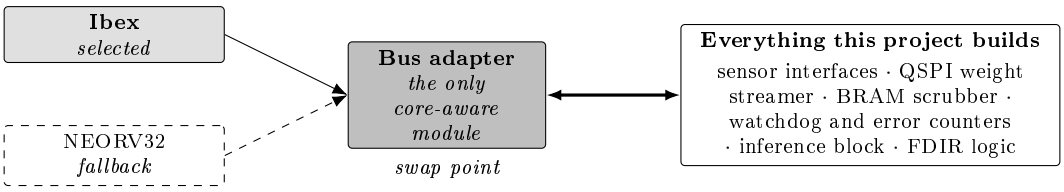


Figure 5: The bus adapter confines core-specific logic to one module. Switching base cores means rewriting the adapter, not the system: an estimated few days rather than the months a full re-integration would cost. This is what makes the NEORV32 fallback in Table 4 a genuine option rather than a formality.

Two consequences follow. The fallback becomes real: if Ibex bring-up stalls, the switch costs days, and the only loss is configuration C3 of Figure 7, since lockstep would then have to be built rather than enabled. And the port itself becomes a small result — running the identical workload and fault campaign on two different cores through one adapter is a cleaner comparison than the literature usually manages, because everything except the core is held constant.

I want to be clear that this is the *only* hedge being taken. A hybrid design carrying both cores was considered and rejected: their bus interfaces and memory maps differ enough that it would mean building the integration twice for no experimental gain.

Deliverable of WP1. Ibex running on the Boolean Board with `SecureIbex` *disabled*, inside an SoC of my own construction, booting a real-time operating system and executing the motion, thermal and extrusion control loops against a simulated plant. No protection yet — that is WP2. This build is configuration C0 of Figure 7, the reference against which every later measurement is taken, so it is a deliverable in its own right rather than an intermediate step.

7.5 WP2 — Make the SoC radiation-tolerant (months 4–7)

WP2 adds protection to the WP1 baseline and measures what each addition costs. Every mechanism below can be switched on or off when the design is built, which is what allows a fair comparison: the same processor, the same workload, one variable at a time.

Protection mechanisms. The work divides cleanly into configuration and construction.

Enabled in Ibex and characterised: dual-core lockstep with temporal diversity; register-file ECC with write-enable glitch detection and one-hot address decoding; instruction-cache ECC; hardened program counter; shadow CSRs on critical control state. Each is a build-time parameter, which is precisely what makes the controlled comparison possible.

Built by this project: periodic scrubbing of idle block RAM; TMR with voting on peripheral control finite state machines, which sit outside Ibex's protection domain; staged watchdog escalation from task through application and core to system reset; error counters that convert Ibex's alert lines into a telemetry time series the FDIR layer can reason over [6]; bus parity, timeout and memory protection; and redundancy on the inference path, which no existing mechanism covers.

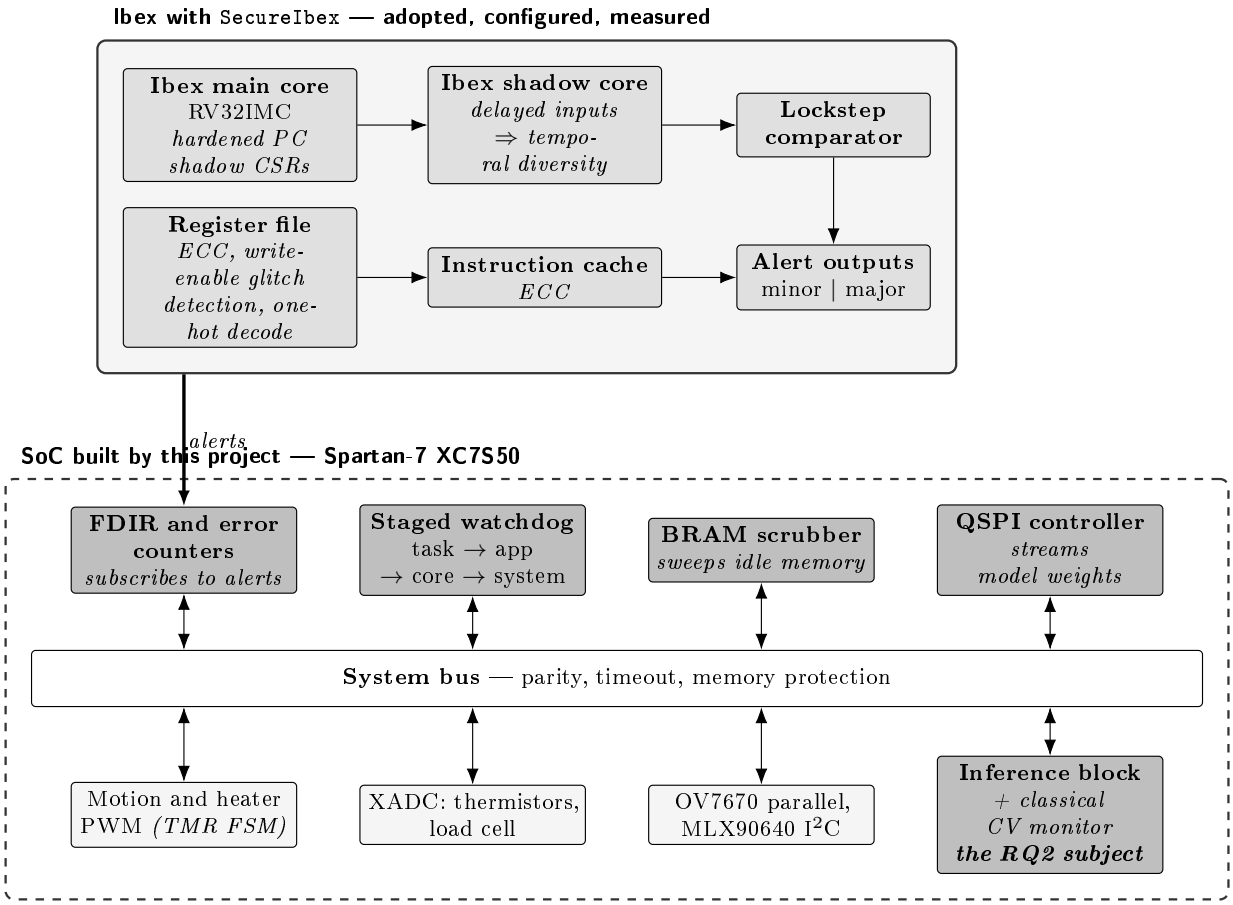


Figure 6: System architecture, drawn to separate what is adopted from what is built. The solid box is Ibex with `SecureIbex` enabled: lockstep with temporal diversity, register-file and cache ECC, hardened PC and shadow CSRs, all existing and all exposed through two alert lines. The dashed box is the SoC this project builds around it, including every mechanism Ibex does not provide — scrubbing, watchdog escalation, error-counter telemetry, TMR on peripheral control logic, and the inference path whose corruption is the subject of RQ2. Ibex supplies the signalling layer; the contribution is what subscribes to it.

Verification. Four layers of checking, from cheapest to strongest. First, the official RISC-V test suite, run through RISCOF, to confirm the processor obeys the instruction set correctly. Second, randomly generated programs from `riscv-dv`, run simultaneously on my design and on Spike, a trusted software model of RISC-V, so that any disagreement between the two is caught automatically. Third, self-checking testbenches for the peripherals and memory, with written targets for how much of the design each test must exercise. Fourth, where the tools allow it, mathematical proof that the small critical blocks — the voters and the error-correction encoders — are correct for every possible input, not just the ones I thought to test.

Target device. Two different FPGAs will be used, for two different reasons.

The **fault-injection platform** is the **RealDigital Boolean Board**, carrying an AMD

Spartan-7 XC7S50-CSGA324, already held by the Department. Being SRAM-based, its configuration memory *can* be corrupted — normally a drawback, but exactly what is needed here, because it allows faults to be injected deliberately, and it is the standard way this kind of testing is done [5,23]. Two properties make this specific device a good fit rather than merely an available one. First, AMD's **Soft Error Mitigation (SEM) IP core** supports Spartan-7 [33]: it uses the device's ICAP and FRAME_ECC primitives to detect, correct and classify configuration-memory upsets, and — critically for this project — it includes a built-in error *injection* feature intended precisely to evaluate SEU mitigation without beam time. The configuration-memory half of WP5 therefore rests on a free, pre-verified, vendor-supported mechanism rather than a harness I would have to write and then justify. Second, the part is comfortably larger than the Artix-7 XC7A35T that would otherwise have been bought (Table 6), which matters for the reason given below.

The board's peripheral set happens to suit the work well. Four Pmod ports across two 30-pin connectors expose 44 FPGA I/O pins, which is enough for the entire sensor suite over I^2C and SPI. An on-board XADC reads the thermistors and load cell without external conversion. The HDMI output can display live layer imagery and classifier verdicts, which makes the system demonstrable at review gates without a host PC. The sixteen slide switches and LEDs are a convenient front panel for selecting hardening configurations and triggering fault injection by hand during bring-up.

One real constraint: no external DRAM. The Boolean Board carries 128 Mbit of QSPI flash but *no* DDR3, unlike the otherwise similar Urbana and Arty S7 boards. Everything the processor executes must therefore live in on-chip block RAM — 75 tiles holding 300 kB of data (the quoted 2.7 Mbit includes parity bits that a byte- or word-wide memory cannot use) — shared between program code, stack, frame buffers and neural-network weights. This is a genuine limit and it shapes three decisions:

- **It settles the base-core choice.** An application-class core such as CVA6 expects external DRAM and is simply not an option here. NEORV32 is designed for exactly this situation: a microcontroller-class SoC running from on-chip memory. What was a preference in Table 4 is now close to a requirement, which removes a decision from month 2 rather than adding one.
- **Weights stream from QSPI.** 128 Mbit is 16 MB — ample for a quantised model. Weights are stored in flash and streamed layer by layer into block RAM during inference, rather than being resident. This is the standard approach for embedded inference on memory-constrained parts.
- **It caps the inspection resolution.** At 320×240 greyscale a frame is 75 kB. One frame, 128 kB of code and 64 kB of data and stack fit in the 300 kB, with 32 kB left for peripheral buffers. Double-buffering would not fit, so frames are processed as they arrive; higher resolution would not fit either. This is an acceptable limit — published FFF defect classifiers operate at comparable resolutions [4] — but it is a limit, and it is recorded as risk R12 rather than discovered later.

The camera is chosen to match: an OV7670 with an 8-bit parallel interface, which maps onto a 30-pin Pmod connector directly. A MIPI CSI-2 module such as the Raspberry Pi camera would need a bridge the Boolean Board does not have.

The **flight-representative target** remains a flash-based FPGA (Microchip IGLOO2, SmartFusion2 or PolarFire), which stores its configuration in flash that radiation cannot flip, so an entire category of failure disappears. That is exactly why such devices are used in space, and the comparison between the two platforms is part of the result.

Why the device size matters to the hypothesis. H1 predicts that selective hardening beats full triple modular redundancy on a reliability-per-resource basis. On this device that stops being an abstract claim and becomes a physical constraint. Taking the published figure of roughly 5.6×

Table 6: Resources of the available device against the alternative it replaces.

	XC7S50 *(in the lab)*	**XC7A35T** *(Basys 3)*	
Logic cells	52,160	33,280	+57%
6-input LUTs	32,600	20,800	+57%
Flip-flops	65,200	41,600	+57%
Block RAM	2.7 Mbit	1.8 Mbit	+50%
DSP slices	120	90	useful for quantised inference
External DRAM	none	none	forces an MCU-class core
QSPI flash	128 Mbit	32 Mbit	holds streamed model weights

resource cost for full TMR [5], a modest RISC-V core occupying roughly 5,000 LUTs unprotected would need roughly 28,000 LUTs fully triplicated — against the 32,600 available. Full TMR of the processor alone would therefore consume most of the fabric, leaving little for the inference block and peripherals, whereas selective hardening should fit comfortably. The device does not merely permit the experiment; it makes the trade-off that RQ1 asks about immediately visible, and it is honest to note this is a constraint I will be working against rather than around.

The configuration ladder — what is actually measured. Because every mechanism is a build-time parameter, WP2 reduces to building the same processor several times and measuring the difference. Figure 7 defines the ladder. Each rung is a synthesised bitstream run against the identical control workload and the identical pre-registered fault schedule, so the comparison isolates one variable at a time.

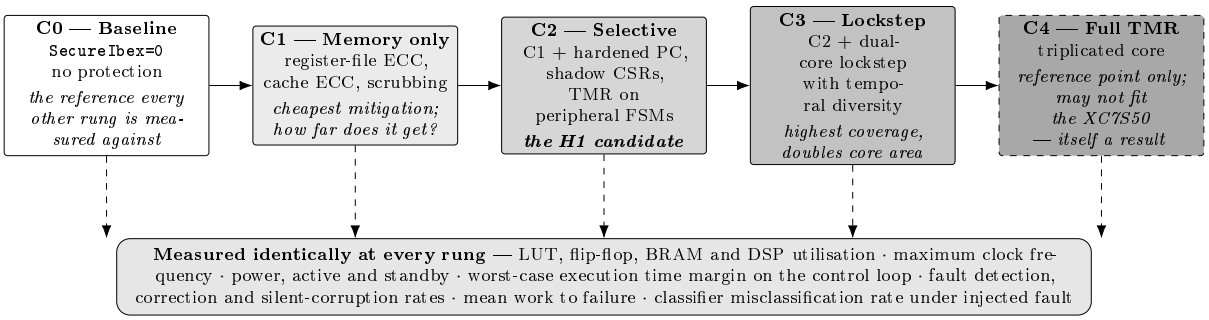


Figure 7: The configuration ladder. Ibex's parameters make the independent variable genuinely independent: the same RTL, the same workload and the same fault schedule, with one protection mechanism added per rung. H1 predicts that C2 approaches C3's mean work to failure at substantially lower cost, and that C4 either fails to fit or violates the control loop's real-time deadlines — in which case that is a finding rather than a failure.

Deliverable of WP2. Five synthesised configurations and the table comparing them: resource cost, speed penalty, power, real-time margin, and how many injected faults each catches. This answers the first half of RQ1 on its own, before the printer exists — which is why WP1 and WP2 together constitute the descope floor of Section 11.

7.6 WP3 — Design and build the printer (months 3–7)

This runs alongside WP1 and WP2 because it involves ordering parts and waiting for them, and that waiting should overlap with design work rather than follow it.

Design. The machine is modelled completely in SolidWorks first: a CoreXY motion frame in aluminium extrusion, an *evacuable* build chamber with O-ring sealing and electrical and fluid feedthroughs, a liquid-cooled hot end, a temperature-controlled build plate, resistive film wall heaters, a tension-controlled filament path, and mounts for every sensor. Working in CAD before cutting metal matters more than usual here, for two reasons: sensor positions — particularly the camera angles and the thermal array's field of view — have to be right for the imaging to be repeatable and are painful to change afterwards; and every penetration through a vacuum boundary has to be planned rather than added later.

Analysis. The CAD model feeds ANSYS. Structural analysis checks frame stiffness, finds resonances that would blur the layer imagery or shake the extruder, and sizes the chamber wall against one atmosphere of external pressure. Thermal analysis is run in the three configurations of Section 7.2 to set build-plate power, wall temperature and dwell time. This is the step that makes the machine microgravity-representative by design rather than by assertion.

Trade study: conducting heat off a moving hot end. This is the one genuinely open mechanical decision in the build, and it is resolved in month 4, before fabrication, using ANSYS conductance modelling and measured drag on a single-axis rig. Table 7 sets out the options.

Table 7: Options for removing heat from the moving hot end, to be decided in month 4.

Option	Thermal conductance	Cost to the motion system	Failure and contamination risk
Braided copper strap	Moderate; scales with cross-section, so more heat means more stiffness	Adds a position-dependent restoring force; the stiffer the strap, the worse the following error	Very low. No fluid, no wick, no moving parts. Fatigue over many cycles is the only concern
Heat pipe with a compliant section	High effective conductance for its mass	Low if the pipe is mounted along the carriage; bend radius is the constraint	Wick performance degrades if the pipe is flexed repeatedly; heat pipes are designed to be rigid
Fluid loop with a service loop	Highest; this is what the published vacuum-FFF systems use [17]	Tubing drag plus the mass of the coolant	Highest. A leak inside an evacuated chamber is a contamination event, and the pump is a wear item

The expected outcome is a braid as the baseline, with a heat pipe carrying heat along the carriage to the braid attachment point if the conductance proves insufficient. The fluid loop is retained only as a fallback, because a leak into the chamber is the worst failure mode available and a pump reintroduces exactly the wear item the design is trying to remove. The measured following error under each option is reported either way — it is a small result that anyone building a vacuum-capable FFF machine would want.

Build. Fabrication, wiring, and bring-up of the motion and thermal hardware under a conventional controller at atmospheric pressure first, so that the mechanics are known to work before

either the vacuum or the SoC is introduced. Pump-down and leak checking follow, then the pressure sweep of Section 3.5. Only then are the sensors added:

- cameras looking down at each layer and across it at an angle, under lighting I control so images are comparable from print to print;
- an infrared array and thermistors, to see how the part is cooling;
- an accelerometer and a contact microphone, since a clogged nozzle, a jam or a worn belt each make a distinctive sound and vibration;
- an encoder and load cell on the filament, to catch slipping feed and measure how hard the extruder is pushing;
- chamber pressure and fume sensors, to confirm the enclosure is sealed.

ABS will be used for day-to-day development because it is cheap. A small quantity of polycarbonate will be bought for comparison, since it is closer to the materials actually printed on the ISS [1].

Deliverable of WP3. A working, fully instrumented printer producing repeatable parts under a conventional controller, with CAD, analysis results and fabrication drawings documented. Note that this deliverable stands alone: it is useful to the Department as a research testbed regardless of what happens to the rest of the project.

7.7 WP4 — Integrate and test (months 7–9)

WP4 is where the two threads meet: the protected processor from WP2 takes over the printer from WP3. The control laws are designed and validated in MATLAB and Simulink first — plant models identified from measured step responses of the real machine, controllers tuned and discretised, stability margins checked, and the whole loop run in hardware-in-the-loop before the SoC is given authority over the heaters. The Simulink plant model is then kept as the reference that sensor residuals are measured against, which is what makes the process-fault branch of Figure 8 work.

- **The fast loop, on the processor:** deterministic motion, thermal and extrusion control, with worst-case execution time analysis establishing real-time margins.
- **The slow loop, once per layer:** optical and thermal inspection → defect classification → corrective action (flow-rate adjustment, temperature adjustment, layer reprint, retract and purge, abort).
- **Classifier:** a transfer-learned convolutional network, quantised for the SoC, targeting stringing, over- and under-extrusion, warping and layer shift — replicating published systems reporting roughly 90 % to 93 % accuracy [4,21], with the critical addition that this one runs on hardened hardware under injected fault.
- **Redundant classical computer-vision monitor** as an independent check on the learned path — itself an RQ2 mitigation to be evaluated rather than merely an engineering convenience.
- **Hardware safety envelope:** any corrective action capable of producing a hazard (overtemperature, unbounded retry, loss of containment) is inhibited independently of software.

The inspection sensors. The optical channel is an OV7670 on an 8-bit parallel interface. Only the luminance bytes of its YUV422 output are kept, giving a 320×240 greyscale frame of 75 kB; that resolution is set by the block-RAM budget, not by the sensor. One frame is taken per layer, roughly every 30 s to 60 s, from a view looking down on the layer and a view across it at an angle. Illumination comes only from a diffused LED ring at fixed exposure, so images are comparable from print to print. The thermal channel is an MLX90640 array (32×24 pixels, 110° field of view)

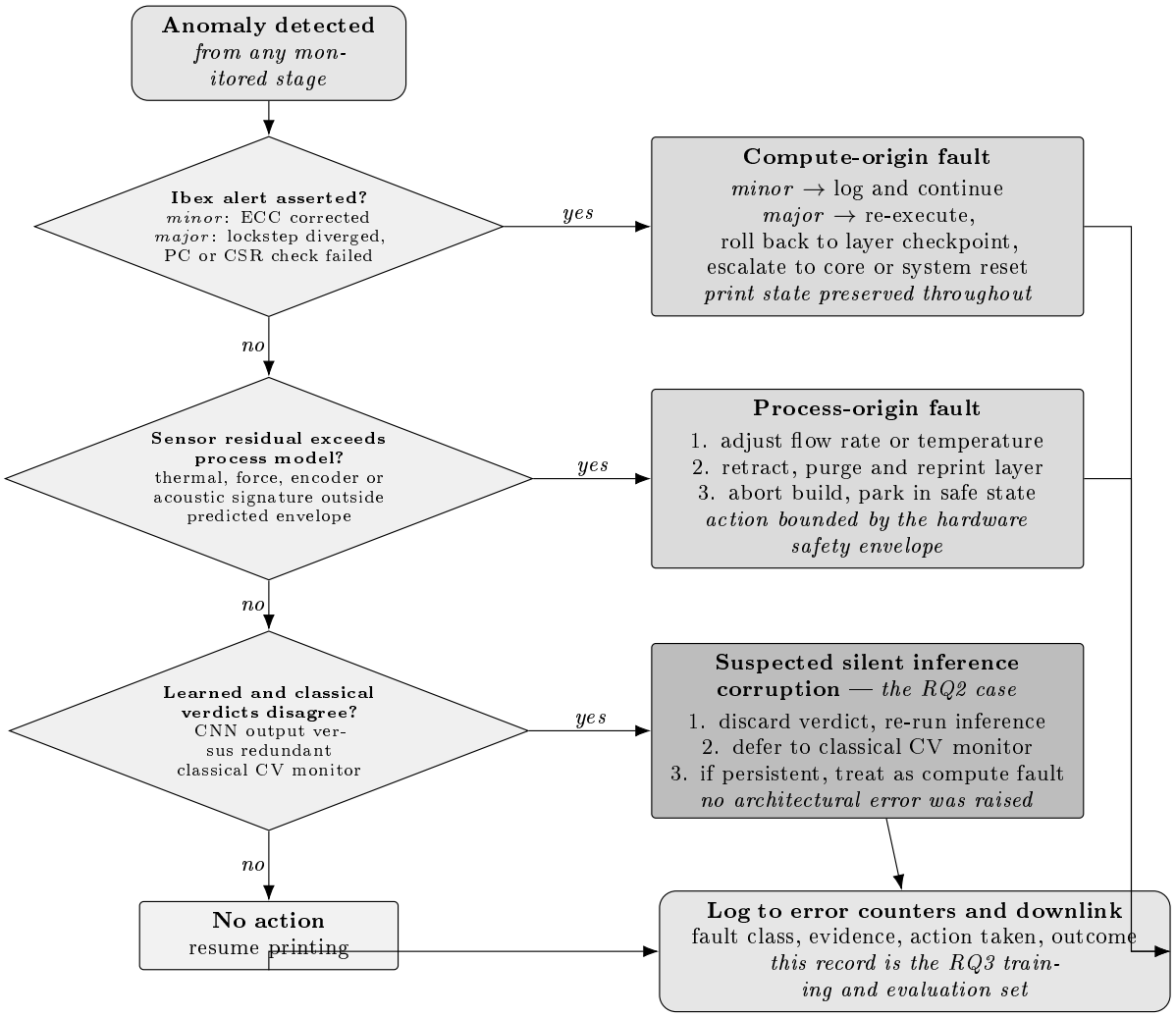


Figure 8: Fault detection, discrimination and recovery. The ordering of the tests is deliberate: an Ibex alert is unambiguous and is therefore checked first, and its minor/major distinction already separates corrected from uncorrectable faults. Process faults, by contrast, must be inferred from sensor residuals against a model. The third test exists only because of Gap 2 — an upset in the inference path raises no error signal and produces no residual anomaly, so the only remaining evidence is disagreement between the learned and classical verdicts. RQ3 asks how reliably the second and third branches can be separated from the first.

on I^2C. The classifier takes the 320×240 frame as input, is quantised to int8, and is streamed from QSPI flash one layer at a time.

Making the inspection hold up in orbit. An inspection system trained and tested in a laboratory can fail in orbit for reasons that have nothing to do with print quality. Each of the conditions in Table 8 is designed for from the start rather than handled afterwards, and each has a ground test that stands in for it. None of this amounts to flight qualification: the OV7670 is a ground prototype, and a flight unit would use a space-qualified image sensor alongside the flash-based FPGA discussed in Section 7.5. Lens contamination from outgassing is flagged at run time but not characterised, for the reason given in Section 3.4.

Table 8: Conditions in orbit that affect the inspection loop, and how each is handled and tested on the ground.

Condition	Effect on inspection	Design response and ground test
Radiation on the image sensor	Hot and dead pixels; corrupted pixel values or lost rows	Training data augmented with hot pixels, dead pixels, bit flips and dropped rows. A dark frame with the LEDs off is taken each layer to map hot pixels before classification
Radiation on weights and processor	Silently wrong classification, with no error signal (Gap 2)	CRC check on the weights before each inference. A corrective action requires agreement with the classical monitor; disagreement pauses the print rather than acting (Figure 8). Measured under fault injection in WP5
No convective cooling	Sensor temperature and image noise rise during long prints	Sensor temperature is logged with every frame. Accuracy is measured on images taken at 1 mbar, not only at ambient pressure
Outgassing and fumes	Haze builds up on the lens over a print	The empty bed is imaged before each print and compared with a reference; a drop in contrast flags the lens for attention
No gravity on the melt	Defects look different: no sagging or drooping of overhangs	Training and test sets include prints made on the side and inverted, following the orientation tests of Section 3.5; accuracy must not depend on which way gravity points
No ambient light	Scene lit only by the onboard source	Already the design case: diffused LED ring at fixed exposure
Launch vibration	Camera pose shifts and focus drifts	Lens focus is locked mechanically and mounts are rigid. Fiducial marks on the bed let the camera pose be re-estimated before each print

The test that matters most is the one Gap 2 points to: with faults injected into the image path, the stored weights and the processor, how often does a wrong corrective action get through? That rate is reported as a result of WP5 alongside the classification accuracy.

Deliverable of WP4. An integrated machine that prints unattended, detects at least four classes of defect and corrects them without intervention, all under the protected processor.

7.8 WP5 — System-level testing (months 9–12)

This constitutes the scientific core of the project.

Compute-fault injection, at two levels within scope.

1. RTL simulation: single- and multiple-bit upsets injected into flip-flops and memory arrays, statistically sampled with reported confidence intervals.

2. FPGA emulation: configuration-memory bit-flipping on the Spartan-7 using the AMD SEM IP core, whose error-injection feature exists precisely to evaluate SEU mitigation without beam time [33]; user-memory and register injection via a shadow-injection harness.

Beam testing (outside the twelve months; the natural continuation). IUAC New Delhi operates the national heavy-ion facility used by ISRO for single-event-effects qualification of microelectronics [14], and allocates beam time to university researchers through a formal proposal process. Approaching IUAC with a completed, characterised design and published emulation results is a far stronger proposition than approaching them with an intention. **The project does not depend upon beam access, and no radiation-hardness claim will be advanced on the basis of emulation alone**; results are reported as fault-injection cross-section proxies with the limitation stated explicitly.

Process-fault injection. Reproducible physical faults: filament runout, induced partial nozzle clog, commanded layer shift, adhesion loss, feed slip and thermal excursion.

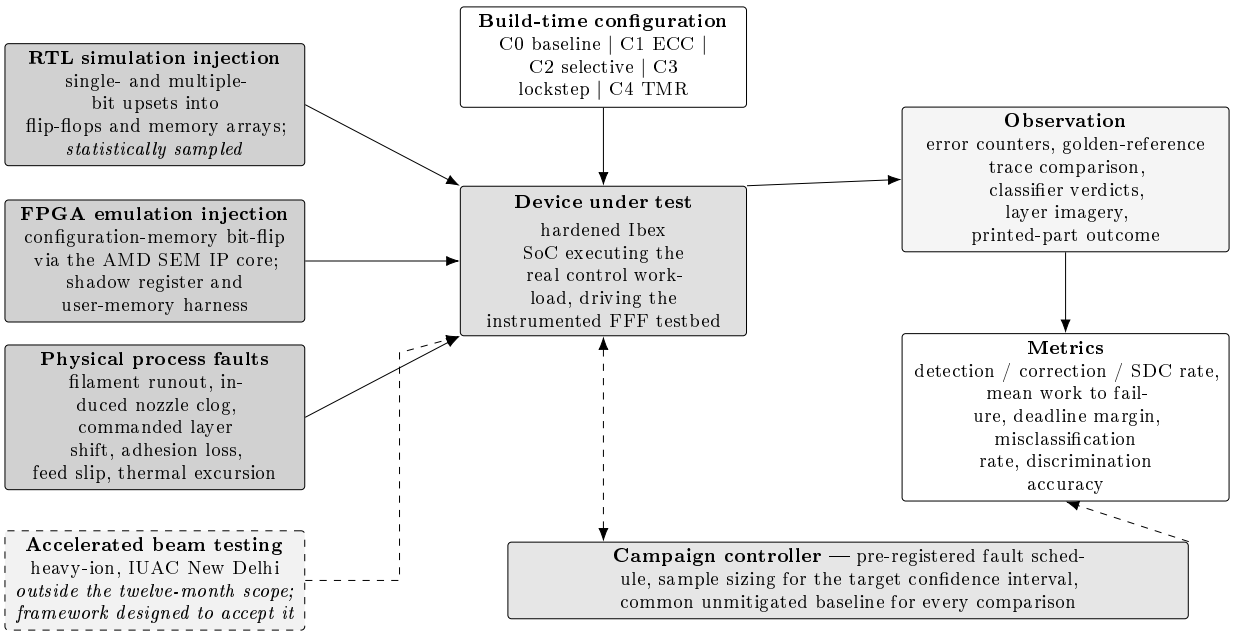


Figure 9: Fault-injection framework. Three injection routes drive a single device under test whose mitigations are selected at build time, so that every configuration is compared against a common unmitigated baseline under an identical, pre-registered fault schedule. The framework is deliberately structured so that accelerated beam testing can be added later as a fourth injection route without altering the measurement or analysis path.

Experimental rigour. Every configuration comparison is executed against a common baseline build with all mitigations disabled. Sample sizes are determined in advance for the target confidence interval, and the fault-injection schedule is fixed before the campaign commences. Ground control specimens are printed from a common feedstock lot under identical documented settings — the methodological discipline that rendered the NASA Phase I and Phase II results interpretable [2].

8 Power

Power is treated separately from thermal design because the two domains involved differ by two orders of magnitude and are limited by different things. Confusing them is how in-space manu-

facturing concepts end up proposing printers that no spacecraft could run.

8.1 Two domains, deliberately kept apart

Table 9: The two power domains. They are not comparable, and the design consequences run in opposite directions.

	Domain A: the controller	**Domain B: the manufacturing unit**
What it is	Hardened RISC-V SoC on FPGA	Heaters, motion, feedstock handling
Bench figure	2 W to 5 W	200 W to 250 W steady; ≈1 kW peak during warm-up
Dominated by	Logic switching and memory	Resistive heating, overwhelmingly
Limited by	*Fabric*, not power	*Power*, absolutely
Hardening effect	Multiplies this domain	Unaffected

The consequence, which is not obvious until the numbers are done. The manufacturing unit draws roughly fifty to a hundred times the controller. Tripling the processor's power through full triple modular redundancy therefore moves the *system* power budget by well under one percent. **Power is not a reason to hold back on hardening the processor; fabric is.** This makes the RQ1 trade-off asymmetric in a way the literature does not discuss, and it is worth stating plainly because it cuts partly against H1: the resource argument for selective hardening survives, since the XC7S50 has 32,600 LUTs and full TMR would consume most of them, but the power argument for it does not. Both are measured; they are expected to disagree; and the disagreement is a result.

8.2 Domain A: powering and measuring the controller

Measurement. Power is added to the metrics recorded at every rung of the configuration ladder (Figure 7), by two independent means: Vivado's vector-based `report_power` driven by simulation activity from the real control workload, and an inline USB power meter on the Boolean Board's supply. The absolute figures from an FPGA are not the figures a flight ASIC would show — FPGA fabric is far less efficient than dedicated silicon — so *only the differential across configurations is claimed.* Everything except the bitstream is held constant, which makes the comparison valid even though the absolute number is not transferable.

A prediction worth recording in advance. The power penalty of hardening should be *sub-linear* in the area penalty, because the Ibex core is a small fraction of total system power once memory, I/O and the inference block are counted. Lockstep doubles the core but should move system power far less than double. If this holds, it is a useful result for anyone sizing a hardened controller, and it is not something the existing literature reports, which quantifies hardening in LUTs and megahertz rather than watts.

Powering the chip in flight. A spacecraft supplies an unregulated battery bus, typically 12 V to 28 V, from which point-of-load converters derive the core, I/O and memory rails. Two points matter and both are noted as scope boundaries rather than work items. First, the regulators must themselves be radiation-tolerant: hardening the processor achieves nothing if a single-event latch-up in a point-of-load converter destroys the board. Second, latch-up protection — current

limiting with autonomous power cycling — is a board-level function, not an RTL one, and sits outside this project. Both are recorded here so that the boundary of what is being claimed is explicit.

8.3 Domain B: powering the manufacturing unit

Table 10: Power budget, separating the ground testbed from what a flight unit would draw. The distinction matters: the pump exists only to manufacture, on the ground, an environment that orbit supplies free.

Load	**Ground, in air**	**Ground, at** 1 mbar	**Flight**	**Note**
Heated bed	100 W	40 W	40 W	Hold power; ≈360 W during warm-up
Chamber heater	80 W	25 W	25 W	No convective loss below ≈10 mbar
Hotend	40 W	40 W	40 W	Unchanged
Steppers (outside chamber)	32 W	32 W	32 W	Externally mounted; see below
Controller and sensors	10 W	10 W	10 W	Domain A
Vacuum pump	—	370 W	—	**Ground test equipment only**
Steady	≈260 W	≈150 W	≈150 W	Pump off once the chamber is sealed
Peak (warm-up)	≈600 W	≈970 W	≈600 W	Ground peak includes simultaneous pump-down

These figures are estimates, and are stated as such. Every number in Table 10 is a component-rating calculation, not a measurement. Nothing has been built, so the totals carry the usual uncertainty of a paper exercise: heater duty cycles depend on insulation quality and chamber leak rate, and the reduction in hold power under vacuum is a prediction from the thermal argument of Section 3.4 rather than an observation. Measuring the real figures is part of WP3, and the difference between predicted and measured hold power is a small result in its own right.

For a sense of scale, NASA's 3D Print — the first printer flown to the ISS, in 2014 — drew 176 W, massed 20 kg and had a $6 \times 12 \times 6$ cm build volume, running ABS [34]. The estimate here sits in the same band for a machine of broadly similar class, which is weak evidence that the budget is not wildly wrong. It is no more than that: 3D Print operated in cabin air inside the Microgravity Science Glovebox, so it had convective cooling and convective loss, whereas this design has neither. The two thermal regimes are different enough that the comparison establishes an order of magnitude and nothing finer. The later ISS facility, AMF, is installed in an EXPRESS Rack, and the planned multi-material FabLab is constrained to 2 kW peak by rack limits rather than by the printer's own draw [34].

The pump is test equipment, not a subsystem. This is worth stating plainly because it inverts the usual relationship between a testbed and the thing it represents. On the ground, roughly 370 W and a rotary-vane pump are needed to *create* an environment that an unpressurised orbital platform provides permanently and for nothing — and provides far better, since ambient pressure in low Earth orbit is orders of magnitude below the 1 mbar this chamber targets. **In the flight configuration the pump is deleted and replaced by a vent valve.** The consequences are all favourable:

- **Peak power falls** from ≈970 W to ≈600 W, since pump-down never coincides with warm-up.
- **Energy per part falls.** Ground operation pays a pump-down cost of roughly 0.09 kW h per cycle that flight does not, so the ground figures in Table 11 are conservative.
- **A wear item disappears.** A rotary-vane pump has bearings, vanes and oil — exactly the class of component this design has been removing elsewhere, for a machine meant to run unattended for weeks.
- **A vibration source disappears.** The pump is switched off during printing on the ground anyway, precisely because it would inject broadband noise into the accelerometer and acoustic channels used for clog detection.

Even the enclosure changes role. On the ground it is a pressure vessel holding one atmosphere of differential. In flight, on an unpressurised platform, it holds no differential at all and exists only for contamination control and part retention, so it becomes substantially lighter.

The honest consequence: this thermal design assumes an unpressurised platform. If the unit were instead flown inside a pressurised module such as the ISS, cabin air would be present, convection would return, and either the chamber would have to be actively evacuated — reintroducing a pump — or the thermal architecture of Section 3.4 would have to be redesigned around forced convection, as the existing ISS printers are. The design here follows from the stated target of Section 2.2: an autonomous unit operating where crew are absent. That is a deliberate commitment rather than an oversight, and it should be visible to the reader.

Vacuum cuts the running cost. The same absence of convection that complicates cooling also removes the dominant heat-loss path, so holding the chamber at temperature costs roughly a third of what it does in air. Warm-up energy is unchanged. The consequence is a scheduling one: **long continuous prints are far more energy-efficient than many short ones**, because warm-up is amortised. This reinforces the point already made about layer dwell time — thermal and power management collapse into the same scheduling problem for the planner running on the SoC.

Motors must sit outside the chamber. Stepper motors and their drivers are rated on the assumption of convective cooling. At 1 mbar they overheat — the same physics that motivates the entire thermal argument, now working against the design. The CoreXY frame is therefore laid out with motors mounted externally, driving the stage through belts and rotary feedthroughs. This is a constraint on the SolidWorks work of WP3 and is far cheaper to honour in CAD than to discover after fabrication.

Thermoelectric wall cooling is dropped. An earlier version of this design specified thermoelectric modules for commanded wall temperature. For ABS and polycarbonate the chamber wants to be *hot* to prevent warping, so the cooling mode would almost never be used. Resistive heating with passive loss to ambient achieves the same control at a fraction of the power, with no H-bridge and no additional failure mode.

8.4 What this would mean in orbit

Sizing a self-powered unit. Supporting 150 W orbit-average in low Earth orbit, where roughly 35 % of each ≈92 min orbit is spent in eclipse, requires about 230 W generated while sunlit. At the solar constant of 1361 W/m^2 and 30 % triple-junction cell efficiency, that is roughly 0.6 m^2 of cell area before pointing losses, degradation and packing — realistically closer to 1 m^2. Carrying the load through a 35 min eclipse needs on the order of 90 W h of usable battery, some 1 kg to 2 kg

Table 11: Feasibility of a 150 W manufacturing unit against platform power, on the estimated figures of Table 10. One part is taken as ≈1 kW h, being a three-hour print plus warm-up. The conclusions are robust to a factor-of-two error in the estimate; the boundary between the marginal and comfortable rows is not.

Platform	**Orbit-average power**	**Energy per day**	**Verdict**
3U CubeSat, body-mounted cells	5 W to 10 W	≈0.2 kW h	Infeasible: one part would consume the entire spacecraft's output for four to eight days
6U–12U with deployable arrays	20 W to 50 W	0.5 kW h to 1.2 kW h	Marginal; viable only for very small parts with aggressive duty cycling
ISS internal payload	≈500 W per locker	continuous	Comfortable — and precisely why the only operational in-space printer is on the ISS
Commercial free-flyer	1 kW to 5 kW	continuous	Comfortable; the Phase 7 architecture

of cells at typical depth of discharge. A manufacturing payload is therefore not a small addition to a spacecraft; it sizes the spacecraft.

The design consequence, and why it belongs in this proposal. Printing must be synchronised to the orbit: build during the sunlit pass, and either hold the chamber warm through eclipse at a cost in battery, or accept a thermal cycle that damages layer adhesion. That is a decision the onboard planner has to make, against a power forecast, a thermal model and a print queue — and it is the same planner already responsible for regulating interlayer temperature through dwell time (Section 3.4). **Power scheduling and thermal scheduling are one problem, and it lands on the controller this project is building.** Implementing an orbit-synchronised planner is beyond the twelve months, but the control architecture is structured so that the outer loop takes a resource forecast as an input rather than assuming unlimited power, which is what makes it extensible to that case rather than a dead end.

9 Work Plan and Milestones

The schedule is built around two constraints. First, the critical path runs WP0 → WP1 → WP2 → WP5: the system-level campaign cannot begin until a protected processor exists, and WP1 and WP2 alone are sufficient to answer RQ1 even if everything downstream fails. Second, WP3 — building the printer — is started in month 3 and run alongside the processor work rather than after it, because it involves ordering parts and waiting for them. Overlapping the two means a delivery delay costs slack instead of schedule. The two threads are developed against an interface frozen at the end of month 2: a written specification of the commands and telemetry passing between the SoC and the printer, so that neither side blocks the other.

Each work package ends in something concrete and independently useful — a verified processor, a measured protection comparison, a working instrumented printer, an integrated closed-loop machine, a characterisation dataset — so that the project degrades gracefully rather than producing nothing if it stops early.

Review gates fall quarterly (M2, M5, M8, M11). The month 2 gate is different in kind from the others: it is a genuine termination point, not a progress review, and is discussed in Section 7.3.

Contingency. Should the printer not be under SoC control by month 8, WP5 is descoped to compute-fault characterisation against a hardware-in-the-loop simulated control workload. This

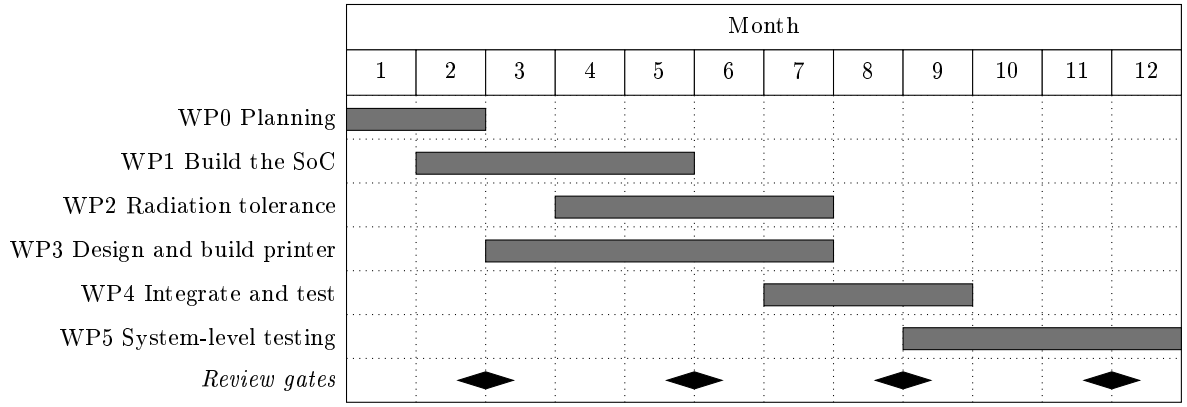


Figure 10: Twelve-month work package schedule with quarterly review gates. Each package terminates in a concrete deliverable before the next depends on it.

Table 12: Milestones and evidence.

Month	**Milestone**	**Evidence**
M2	**Go/no-go gate (WP0).** Ibex running the compliance suite on the Boolean Board; bus adapter defined; printer design frozen in CAD; SoC–printer interface specified	CI report; live demonstration; adapter specification; SolidWorks model
M5	**WP1 complete.** Verified unprotected SoC running the control workload against a simulated plant; ANSYS structural and thermal analysis complete	Live demonstration; analysis report
M7	**WP2 and WP3 complete.** Protected SoC characterised against the baseline; printer built, instrumented and printing repeatably under a conventional controller	Resource, timing and fault-injection report; working machine
M8	Review gate; SoC drives the printer's motion and thermal loops	Live demonstration
M9	**WP4 complete.** Closed-loop defect detection and correction operational; fault-injection framework validated	Accuracy figures; open-source release
M11	Review gate; system campaign complete, RQ1 and RQ2 answered; first paper submitted	Results report; dataset; submission record
M12	**WP5 complete.** Report submitted; second paper submitted	Documents; artefact release

continues to address RQ1 and RQ2 and remains publishable. The descope is declared in advance so that the project may fail gracefully rather than late.

Deliverable of WP5. The answers to RQ1 and RQ2 with stated confidence intervals, a released dataset and fault-injection framework, and the submitted papers.

10 Expected Outcomes and Contributions

Scientific contributions.

1. First characterisation of RISC-V hardening overhead against a hard real-time additive manufacturing control workload, in place of synthetic benchmarks.
2. First quantification of single-event-upset propagation through an ML-driven manufacturing control decision, with evaluation of redundancy strategies for that path.
3. A methodology for establishing the microgravity representativeness of a ground testbed by means of orientation invariance and process-setting control.
4. *(Extension)* A unified compute/process fault taxonomy and discrimination method for autonomous manufacturing systems.

Artefacts, all released open-source.

5. Hardened, configurable RISC-V SoC RTL with independently selectable mitigations.
6. Fault-injection framework spanning RTL simulation, FPGA emulation and physical process faults, reusable by any group undertaking comparable work.
7. Labelled defect dataset with paired hardware error-counter telemetry; no comparable dataset presently exists.

Table 13: Planned publications and target venues.

Output	**Target venue**
Hardening architecture and fault-injection results	IEEE International Symposium on VLSI Design and Test (VDAT), VLSI Design Conference, RADECS, or *Microelectronics Reliability*
Closed-loop control under fault; dataset release	IEEE Aerospace Conference, *Journal of Manufacturing Processes*, or *Additive Manufacturing*
Review and gap analysis *(if time permits)*	*Acta Astronautica* or *IEEE Access*

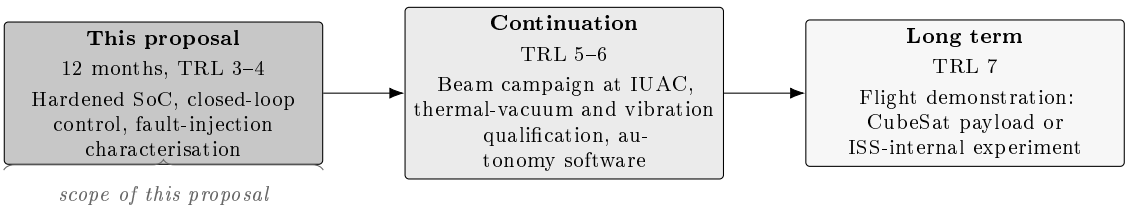


Figure 11: Position of the proposed work within a longer trajectory. Only the first stage is being proposed; it is scoped to stand alone, and produces the evidence package required to pursue the second.

Trajectory. This project resides at TRL 3–4 as defined in NPR 7123.1C [35]. It constitutes the first phase of a longer roadmap encompassing system architecture, autonomy software, environmental qualification and eventual flight demonstration; it is nonetheless deliberately scoped to stand alone. Should the work proceed no further, it remains a complete and novel contribution. Should it continue, it constitutes the evidence package for a beam-time proposal to IUAC and for subsequent funding applications.

11 Resources Requested

11.1 Equipment and consumables

The budget is presented as four tranches tied to the review gates, not as a single request. Each tranche is released only if the previous gate is passed, so the Department's exposure at any point is bounded by what has already been demonstrated. If the project is stopped at any gate, the spend to that point has produced something the Department keeps.

Table 14: Phased budget. Each tranche is contingent on passing the preceding gate.

Tranche	Released	Amt. (Rs.)	What it buys, and what exists if the project stops here
T1	At approval	3,000	Sensors for characterising the existing printer. Deliverable: instrumented baseline and a labelled defect dataset.
T2	M2 gate passed	32,100	Printer build and remaining sensors. Deliverable: a working custom instrumented printer, retained by the Department as a research testbed.
T3	M5 gate passed	20,900	Vacuum chamber and thermal control. Deliverable: a convection-free process chamber and the pressure-sweep result.
T4	M7 gate passed	11,500	PolarFire Discovery Kit for the flight-representative comparison. Deliverable: the flash-versus-SRAM measurement.
Total		**67,200**	≈US$790

Two decisions cut this well below what the earlier version of this proposal assumed. First, the Department already holds a RealDigital Boolean Board (Spartan-7 XC7S50), so no SRAM-based FPGA is bought; the part is larger than the Artix-7 board that would otherwise have been specified, and its Pmod ports and on-board XADC remove the need for a separate sensor-interface board at this stage (Section 7.5). Second, and more significantly, the vacuum requirement was derived rather than assumed: 1 mbar suppresses convection with an order of magnitude of margin (Section 3.4), which is reached by a single-stage rotary-vane pump at roughly Rs. 8,000 rather than the Rs. 22,000 two-stage pump needed for 10^{-3} mbar, and needs only an analogue gauge. Specifying more vacuum than the physics requires would have cost about a fifth of the budget for no experimental gain.

Where an item is commonly found in university laboratories it is marked ‡, with the borrowed-equipment total given at the foot of Table 15. Realistically, if a rotary-vane pump can be borrowed from Physics or Chemistry and a dead printer scavenged for motion hardware, the cash requirement falls to roughly Rs. 51,700.

Table 15: Bill of materials, grouped by tranche. Item names link to the supplier page. Items marked ‡ are commonly available in university laboratories and may be borrowed rather than bought.

Item, part number and supplier	Est. (Rs.)	Purpose
Already held — no cost		
RealDigital Boolean Board, Spartan-7 XC7S50-CSGA324 *(Department)*	—	Fault injection via the SEM IP core; 52,160 logic cells; 4 Pmod ports, XADC, HDMI out
Vivado, Libero, Verilator, RISCOF *(free editions)*	—	FPGA and verification toolchain
SolidWorks, ANSYS, MATLAB *(Institute licences)*	—	CAD, analysis and control design
T1 — At approval (Rs. 3,000)		
OV7670 camera module (8-bit parallel, Pmod-compatible) and LED ring with diffuser *(Robu.in)*	1,400	Layer-wise inspection under controlled lighting; parallel interface avoids the MIPI bridge the board lacks
ADXL345 accelerometer; piezo contact disc *(Robu.in)*	900	Nozzle clog, jam and wear detection
AS5600 magnetic encoder; NTC thermistors *(Robu.in)*	700	Feed slip; temperature
T2 — After the M2 gate (Rs. 32,100)		
Aluminium extrusion, brackets, fasteners *(local)*	4,500	CoreXY frame; stiffness set by ANSYS
Linear rails, belts, pulleys, idlers‡ *(Robu.in)*	4,000	Motion stage
4× NEMA17 steppers with TMC2209 drivers‡ *(Robu.in)*	3,500	Motion and extrusion axes
All-metal hotend (300°C rated), Bowden extruder *(Robu.in)*	3,500	High-temperature capability; Bowden keeps the extruder motor outside the chamber
Heated bed; 24 V 500 W PSU; MOSFET and solid-state relay switching; opto-isolators *(Robu.in)*	4,500	Bed adhesion and power distribution; isolation between FPGA and power stages
Bimetallic thermal cutout, emergency stop, fused mains inlet *(local)*	1,200	Hardware safety chain in series with the heater supply, independent of the processor (Section 8)
INA226 current and power monitor *(Robu.in)*	400	Differential power measurement across ladder configurations
Waveshare MLX90640 IR array, 32×24 px, I^2C *(Robu.in)*	4,500	Layer and bead thermal field
1 kg load cell with HX711 amplifier; 2× MAX6675 thermocouples *(Robu.in)*	2,000	Extrusion force; nozzle temperature

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Table 15 continued from previous page

Item, part number and supplier	Est. (Rs.)	Purpose
BME280; MQ-135 VOC sensor *(Robu.in)*	900	Containment and off-gassing
ABS filament, 2 kg *(local)*	2,000	Development lots
Perfboard, connectors, wiring, level shifters *(Robu.in)*	1,100	Sensor interface, prototyped before PCB
T3 — After the M5 gate (Rs. 20,900)		
Chamber: stainless stock pot or acrylic cylinder, O-ring, acrylic lid, viewport *(local)*	4,000	Build volume; the standard resin-degassing chamber construction
Single-stage rotary-vane pump, 1 mbar class‡ *(local / IndiaMART)*	8,000	**Ground test equipment only** — creates on the bench the vacuum orbit provides free; deleted in any flight configuration (Section 8)
Analogue vacuum gauge and hose *(local)*	800	Pressure as an experimental variable
Electrical and fluid feedthroughs *(local)*	2,500	Sensor, motor and coolant penetrations
Copper heat pipes (sintered wick), braided copper straps *(Robu.in)*	1,200	Heat off the moving hot end; no pump, no moving parts
High-emissivity coating; polished foil; aluminised MLI film *(local)*	2,000	Wall emissivity as a design variable; insulation
G10 / PEEK standoffs, thermal interface material *(Robu.in)*	1,200	Preventing heat leak into the frame
Silicone film heaters and PT100 sensors for the chamber wall *(Robu.in)*	1,400	Commanded wall temperature; resistive rather than thermoelectric, since the chamber is always heated
Polycarbonate filament, 1 kg *(local)*	3,000	High-temperature comparison material
T4 — After the M7 gate (Rs. 11,500)		
PolarFire SoC Discovery Kit, MPFS-DISCO-KIT (MPFS095T) *(Microchip Direct)*	11,500	Flight-representative target; flash configuration memory is inherently SEU-immune
Total requested	**67,200**	≈US$790
If ‡ items are borrowed	**51,700**	≈US$610
Explicitly deferred		
PEI/ULTEM 9085 filament	—	≈Rs. 60,000 per kg; sought as a donated sample only when flight-representative properties must be measured
Two-layer sensor PCB *(JLCPCB)*	—	≈Rs. 6,000; only once the pinout is settled on perfboard
Liquid cold-end loop	—	≈Rs. 3,000; only if conduction proves insufficient (Table 7)

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Table 15 continued from previous page

Item, part number and supplier	Est. (Rs.)	Purpose
Nexys A7-100T *(DigiKey India)*	—	≈Rs. 40,000; only if the inference block will not fit alongside a hardened core in the XC7S50

Also required: bench instrumentation (oscilloscope, bench supply, logic analyser) and compute for model training and RTL simulation, both from existing laboratory stock.

This is deliberately a low-cost proposal, and the phasing is deliberate too. The novelty resides in the experimental design rather than in expensive equipment, and no tranche is requested before the preceding gate has demonstrated that it is warranted.

11.2 From the supervisor

- **Technical supervision**, particularly on verification methodology and FPGA implementation.
- **Review at the four gates** (M2, M5, M8, M11). I would ask that these be adversarial rather than encouraging, and that the M2 gate be enforced strictly.
- **Publication guidance** and co-authorship.
- **Institutional introductions** where you judge them appropriate: to IUAC New Delhi for eventual beam access, to a materials group for micro-computed-tomography and mechanical characterisation of printed specimens, or to any student satellite activity. Access relationships take longer to establish than hardware.

12 Risk Assessment

The risks below are ordered by their expected effect on the twelve-month deliverable rather than by likelihood alone. Three points are worth stating explicitly.

First, R1 — my own inexperience — is the dominant risk, and it is addressed structurally rather than by assurance: the planning phase of Section 7.3, the month 2 termination gate, and the decisions to adopt rather than author both the base core and the verification infrastructure. If the risk materialises, it does so at month 2 at a cost of two months, not at month 10 at a cost of the project.

Second, the mitigations listed are design decisions already embedded in the plan, not intentions to be acted on if trouble arises. The hardware safety envelope, the redundant classical computer-vision path and the two-tier objective structure all exist in the methodology because of the risks they answer.

Third, R6 is included because a negative result is a legitimate outcome here. If hardening overhead proves incompatible with the control loop's real-time deadlines, that finding answers RQ1 and is publishable; it is recorded as a risk only because it would defeat the engineering goal while advancing the research one.

Table 16: Risk register.

#	Risk	Lik.	Imp.	Mitigation
R1	Candidate lacks prior RTL experience	High	High	Months 1–2 constituted as an explicit preparation phase (Section 7.3) with a strict go/no-go gate; Ibex adopted rather than authored, with its protection mechanisms already implemented and its verification flow already working; a thin bus adapter (Figure 5) keeps the NEORV32 fallback a days-long switch rather than a rebuild
R2	Scope too broad for twelve months	Med	High	Two-tier objectives (Core/Extension, Table 3); declared descope path; each work package ends in a standalone deliverable, so stopping early still leaves something usable; WP1–WP2 and WP3 run in parallel against an interface frozen at month 2
R3	Beam time unobtainable	High	Low	Beam testing is explicitly outside the twelve-month scope; emulation is the baseline method, not a fallback; no claim advanced without data
R4	Building the printer consumes disproportionate effort	High	Med	Standard CoreXY architecture and off-the-shelf motion components — nothing is mechanically novel; SolidWorks and ANSYS work completed before ordering; machine brought up under a conventional controller first, so mechanical and control problems are separated; hard checkpoint at M7
R5	Classifier accuracy insufficient after quantisation	Med	Med	Redundant classical computer-vision path already in the design; a reduced defect-class scope remains an acceptable outcome and still addresses RQ2
R6	Hardening overhead violates real-time deadlines	Low	Low	*This constitutes a finding rather than a failure* — it directly addresses RQ1 and is publishable
R7	Flash-based FPGA board procurement delay	Med	Low	All WP1 and WP2 work proceeds on the Spartan-7 already held by the Department; the flash device is required only for the flight-representative comparison, which is the last measurement rather than the first
R8	Fabrication or machining access unavailable	Med	Med	Design uses extrusion, laser-cut sheet and printed brackets rather than machined parts; the frame can be assembled with hand tools
R9	Vacuum chamber leaks, or pump unavailable	Med	Med	Chamber only needs 1 mbar class, not high vacuum, so elastomer seals suffice; the machine functions at atmospheric pressure throughout, and the pressure sweep is an added measurement rather than a precondition
R10	Moving thermal link cannot carry the hot-end load without degrading positioning accuracy	Med	Med	Resolved by trade study in month 4 before fabrication (Table 7); the pumped fluid loop used by published vacuum-FFF systems is retained as a fallback; measured following error reported either way
R11	Polymer outgassing contaminates the chamber or degrades the print	Med	Low	Reported as manageable in published low-vacuum FFF work [19]; cold trap and carbon getter fitted; ABS and PC both have documented low total mass loss
R12	On-chip memory insufficient for the hardened core plus the inference block	Med	Med	300 kB of usable block RAM is the hard limit; weights stream from the 16 MB QSPI flash rather than residing in BRAM; inspection capped at 320×240, comparable to published FFF classifiers [4]; the Nexys A7-100T remains a costed contingency
R13	Bench load approaching 1 kW during simultaneous warm-up and pump-down	Med	Low	Ground-only condition; the pump is test equipment and is off during printing; warm-up and pump-down are sequenced rather than concurrent; dedicated supply and an independent thermal cutout in the hardware safety envelope; steady load is $\approx$150 W
R14	Motors or drivers overheat inside the vacuum boundary	Low	Med	Design keeps all motors outside the chamber; CoreXY makes this natural since both axis motors are frame-mounted, and a Bowden extruder keeps the fourth outside

13 Rationale

13.1 Why now

RISC-V has rendered processor-hardening research accessible to university groups for the first time: the instruction set is open, high-quality cores are open-source, and verification infrastructure such as `core-v-verif` and RISCOF is freely available [7,30]. Concurrently, in-space manufacturing has advanced from demonstration to national strategy [8, 10], and India's own space programme has a sustained, documented requirement for single-event-effects characterisation of microelectronics [14]. The window during which the intersection of these fields is both open and inexpensive to enter will not remain so indefinitely.

13.2 Why this group

The proposal is directed to this group because its established competence maps onto all three technical pillars of the work, rather than onto one of them.

FPGA-based design is the entire substrate of WP1 and WP2. The project is, in implementation terms, an FPGA design and verification exercise with a manufacturing process attached, and it requires exactly the synthesis, timing-closure and on-device debugging practice that the group teaches and applies.

Image and signal processing on FPGA is the substrate of the inspection path. The layer-wise defect classifier is a constrained image-processing pipeline that must execute within a deterministic time budget on a soft core — a quantisation, dataflow and resource-partitioning problem of precisely the kind the group addresses.

Cryptographic hardware supplies the methodology for WP5, and this is the least obvious but most valuable point of alignment. Fault injection is the shared experimental apparatus of two otherwise separate fields: in cryptographic hardware it is an adversarial technique for differential fault analysis, and in space electronics it is an emulation technique for radiation-induced upset. The injection harness, the campaign statistics and the countermeasure vocabulary — concurrent error detection, spatial and temporal redundancy, parity and residue codes — are substantially the same; only the threat model differs. A group that already reasons about deliberate fault injection is unusually well placed to supervise work on accidental fault injection, and the transfer runs in both directions.

13.3 Why this candidate

I am entering the third year of the B.E. programme in Electronics and Communication Engineering. I have completed the core digital design and electronics coursework and have not yet undertaken substantial independent RTL development; Section 7.3 therefore constitutes the first two months as a structured preparation phase with a strict go/no-go gate at month 2, and Risk R1 records this as the principal risk to the project.

I would rather state that plainly at the outset than have it emerge at month six. What I bring in its place is a defined problem, a survey of the literature sufficient to establish that the problem is unaddressed, a work plan with declared contingencies, and a willingness to be held to the month 2 gate. The project has been scoped on the assumption that the candidate is learning, which is why the base core is adopted rather than authored, the verification infrastructure is adopted rather than built, and the scientific contribution is located in the experimental design rather than in implementation heroics.

13.4 What is requested

Approval to proceed, supervision, laboratory and EDA access, and approximately Rs. 67,200 in hardware and consumables, released in four tranches against the review gates rather than as a single sum, and falling to roughly Rs. 51,700 if a vacuum pump and motion hardware can be

borrowed. In return I commit to a documented, version-controlled and reproducible project with defined review gates and a declared descope path, producing open artefacts and no fewer than two submitted papers.

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